

Figure 17.D.1
The excess demand function for an economy with multiple equilibria.

For a theorist, the best of the all possible worlds is one in which the social situation being analyzed can be formalized in a manner that, on the one hand, is very parsimonious (i.e., uses as inputs only the most indisputable and sturdy traits of the reality being modeled) and, on the other, manages to predict a unique outcome.

The Walrasian model of perfect competition is indeed very parsimonious. Essentially, it attempts to give a complete theoretical account of an economy by using as fundamentals only the list of commodities, the state of the technology, and the preferences and endowments of consumers. However, the other side of the coin is that the theory is not completely deterministic. We shall see in Section 17.F that the uniqueness of equilibria is assured only under special conditions. The Edgeworth box of Figure 15.B.9 and Example 15.B.2 provides a simple illustration that, under the assumptions we have made, multiplicity of equilibria is possible. Figure 17.D.1 represents the excess demand function for good 1 of Example 15.B.2 as a function of p_1 (normalizing to $p_2 = 1$). For another example of multiplicity, see Exercise 17.D.1.

From the theoretical point of view, if uniqueness is not achievable, the next-best property is local uniqueness. We say that an equilibrium price vector is *locally unique*, or *locally isolated*, if we cannot find another (normalized) price vector arbitrarily close to it. If every equilibrium of an economy is locally unique, we say that the *local uniqueness property* holds for the economy. The local uniqueness property is of interest because, if it prevails, then it may not be difficult to complete the theory in any particular application. For example, history may have determined the region where equilibrium lies (it could be the region where equilibrium used to be before a small unanticipated shock to the economy), and in that region we may have a unique equilibrium. In this case, the theory retains its predictive power, albeit only locally. We say that a theory that guarantees the local uniqueness of equilibria is *locally* (as opposed to globally) *determinate*.

The question is then: Is the Walrasian theory locally determinate? The example of Figure 17.D.1 suggests that it is: Every solution to the excess demand equation is locally isolated. But Figures 17.D.2 and 17.D.3 provide a counterexample. The figures depict the offer curves and the excess demand function of an exchange economy with a continuum of Walrasian equilibria. Nonetheless, we should not despair. The situation displayed in Figures 17.D.2 and 17.D.3 has an obvious pathological feel about it; it looks like a coincidence. And indeed, it was shown by Debreu (1970) that such an occurrence is not robust: it can happen only by accident.

We now turn to a formal discussion of these issues. For the sake of concreteness we restrict ourselves, as usual, to the analysis of exchange economies formed by I consumers. Every consumer i is specified by $(\succsim_i, \omega_i)$, where $\succsim_i$ is a continuous,

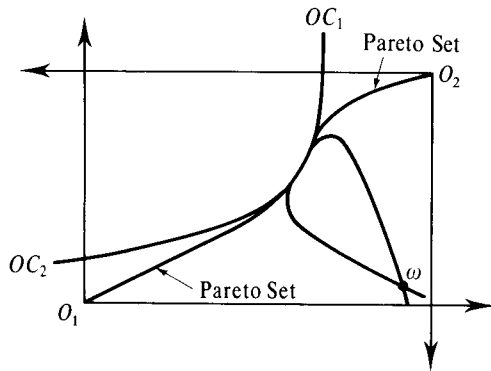


Figure 17.D.2 (left)

A continuum of equilibria is possible: Edgeworth box.

Figure 17.D.3 (right)

A continuum of equilibria is possible: excess demand.

strictly convex, and strongly monotone preference relation on $\mathbb{R}_+^L$ and $\omega_i \gg 0$. As we know, the aggregate excess demand function $z(\cdot)$ then satisfies conditions (i) to (v) of Proposition 17.B.2. We further assume that $z(\cdot)$ is continuously differentiable.¹³

Because we can only hope to determine *relative* prices, we normalize $p_L = 1$ and, as we did in Section 17.B, denote by

$$\hat{z}(p) = (z_1(p), \dots, z_{L-1}(p))$$

the vector of excess demands for the first $L - 1$ goods.¹⁴ A normalized price vector $p = (p_1, \dots, p_{L-1}, 1)$ constitutes a Walrasian equilibrium if and only if it solves the system of $L - 1$ equations in $L - 1$ unknowns:

$$\hat{z}(p) = 0.$$

Regular Economies

It is useful to begin by introducing the concept stated in Definition 17.D.1.

Definition 17.D.1: An equilibrium price vector $p = (p_1, \dots, p_{L-1})$ is *regular* if the $(L - 1) \times (L - 1)$ matrix of price effects $D\hat{z}(p)$ is nonsingular, that is, has rank $L - 1$. If every normalized equilibrium price vector is regular, we say that the *economy is regular*.

In Figure 17.D.1, every equilibrium is regular because the slope of excess demand, $\partial z_1(p_1, 1)/\partial p_1$, is nonzero at every solution. In contrast, none of the equilibria of Figure 17.D.3 are regular because at any equilibrium price vector the slope of the excess demand function is zero. Later in this section we shall argue that, in a sense that we will make precise, “almost every” economy is regular.

The significance of the technical concept of regularity derives from the fact that a regular (normalized) equilibrium price vector is isolated, and a regular economy can only have a finite number of (normalized) price equilibria. This is formally stated in Proposition 17.D.1.

13. In Appendix A to Chapter 3, we discussed conditions for the differentiability of demand functions and therefore of excess demand functions.

14. Nothing in what follows depends on the particular normalization. It can be shown, for example, that if $z(p) = 0$ and the $L \times L$ matrix $Dz(p)$ has rank $L - 1$, then the $(L - 1) \times (L - 1)$ matrix $D\hat{z}(p)$ has rank $L - 1$ whichever good we choose to normalize. Even more, the sign of its determinant is independent of the normalization (see Section M.D of the Mathematical Appendix).

Proposition 17.D.1: Any regular (normalized) equilibrium price vector

$$p = (p_1, \dots, p_{L-1}, 1)$$

is *locally isolated* (or *locally unique*). That is, there is an $\varepsilon > 0$ such that if $p' \neq p$, $p'_L = p_L = 1$, and $\|p' - p\| < \varepsilon$, then $z(p') \neq 0$. Moreover, if the economy is regular, then the number of normalized equilibrium price vectors is finite.

Proof: The local uniqueness of a regular solution is a direct consequence of the inverse function theorem (see Section M.E of the Mathematical Appendix). Intuitively, this is clear enough. For any infinitesimal change in normalized prices, $dp = (dp_1, \dots, dp_{L-1}, 0) \neq 0$, the nonsingularity of $D\hat{z}(p)$ implies that $D\hat{z}(p) dp \neq 0$. Hence, we cannot remain at equilibrium.

Once we know that every equilibrium is locally isolated, the finiteness of the number of equilibria is a consequence of the boundary condition (v) of Proposition 17.B.2 on the excess demand function. Because of this condition (which, recall, follows from the strong monotonicity of preferences), equilibrium is not compatible with relative prices that are arbitrarily close to zero. That is, there is an $r > 0$ such that if $\hat{z}(p) = 0$ and $p_L = 1$, then $1/r < p_\ell < r$ for every ℓ . The continuity of $\hat{z}(\cdot)$ adds to this the fact that the set of equilibrium price vectors is a closed subset of $\mathbb{R}^{L-1}$. But a set that is closed and bounded (i.e., compact) in $\mathbb{R}^{L-1}$ and discrete (i.e., with all its points locally isolated) must necessarily be finite (see Section M.F of the Mathematical Appendix). ■

Our next aim is suggested by reexamining Figure 17.D.1; we see that for a regular economy with two commodities, we can assert more than the finiteness of the number of equilibria. Indeed, the boundary conditions on the excess demand function $z_1(\cdot)$ (excess demand is positive if p_1 is very low and negative if it is very high) necessarily imply that, for a regular economy, first, there is an odd number of equilibria and, second, the slopes of the excess demand function at the equilibrium must alternate between being negative and being positive, starting with negative. If we say that an equilibrium with an associated negative slope of excess demand has an *index* of $+1$ and that one with a positive slope has an *index* of -1 , then, no matter how many equilibria there are, the sum of the indices of the equilibria of a regular economy is always $+1$. With appropriate definitions, it turns out that this invariance of index property also holds in the general case with any number of commodities, where it has some important implications for comparative statics and uniqueness questions.

Let us generalize the definition of the index of a regular equilibrium that we have just suggested for the case $L = 2$ to the case of many commodities.

Definition 17.D.2: Suppose that $p = (p_1, \dots, p_{L-1}, 1)$ is a regular equilibrium of the economy. Then we denote

$$\text{index } p = (-1)^{L-1} \text{sign } |D\hat{z}(p)|,$$

where $|D\hat{z}(p)|$ is the determinant of the $(L-1) \times (L-1)$ matrix $D\hat{z}(p)$.¹⁵

If $L = 2$, then $|D\hat{z}(p)|$ is merely the slope of $z_1(\cdot)$ at p . Hence, we see that for this case the index is $+1$ or -1 according to whether the slope is negative or positive.

15. For any number $\alpha \neq 0$, $\text{sign } \alpha = +1$ or -1 according to whether $\alpha > 0$ or $\alpha < 0$.

A regular economy has a finite number of equilibria (Proposition 17.D.1). Therefore, for a regular economy, the expression

$$\sum_{\{p: z(p)=0, p_L=1\}} \text{index } p$$

makes sense. The next proposition (the *index theorem*) says that the value of this expression is always equal to $+1$.

Proposition 17.D.2: (The Index Theorem) For any regular economy, we have

$$\sum_{\{p: z(p)=0, p_L=1\}} \text{index } p = +1.$$

A brief discussion of why this result is true is given at the end of this section. Here we point out some of its implications and why it is useful and significant. Note, first, that it implies that the number of equilibria of a regular economy is odd.¹⁶ In particular, this number cannot be zero; so the existence of at least one equilibrium is a particular case of the proposition. Second, the index concept provides a classification of equilibria into two types. In a sense, Proposition 17.D.2 tells us that the type with positive index is more fundamental because the presence of at least one equilibrium of positive type is unavoidable. In fact, it is typically the case that any search for well-behaved equilibria (what this means depends on the particular application) can be confined to the positive index equilibria. Third, as we shall see in Section 17.F, the index result has implications for the uniqueness and the multiplicity of equilibria. Fourth, as we shall discuss in Section 17.E, part of the importance of the index theorem is that this is all we can hope to derive without imposing (strong) additional assumptions.

We next proceed to argue that typically (or, in the usual jargon, *generically*) economies are regular. Hence, generically, the solutions to the excess demand equations are locally isolated and finite in number, and the index formula holds.¹⁷

Genericity Analysis

To emphasize the wide scope of the methodology to be presented, we discuss it first in terms of a general system of equations. We then specialize our discussion to the economic problem at hand and spell out its consequences for the excess demand equations.

The essence of genericity analysis rests on counting equations and unknowns. Suppose we have a system of M equations in N unknowns:

$$\begin{aligned} f_1(v_1, \dots, v_N) &= 0, \\ &\vdots \\ f_M(v_1, \dots, v_N) &= 0, \end{aligned} \tag{17.D.1}$$

or, more compactly, $f(v) = 0$. The normal situation should be one in which, with N unknowns and M equations, we have $N - M$ degrees of freedom available for the

16. This result was first shown by Dierker (1972).

17. For advanced treatments on the topic of this section, refer to Balasko (1988) or Mas-Colell (1985).

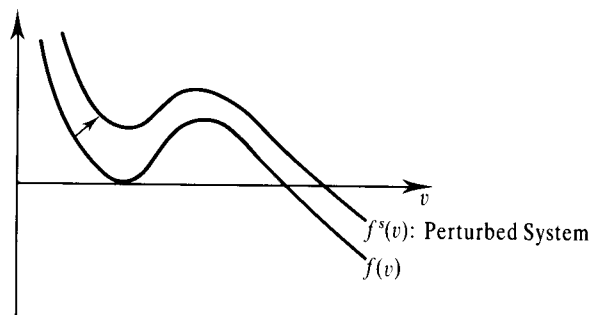


Figure 17.D.4
The regular case is typical.

description of the solution set. In particular, if $M > N$, the system should be *over-determined* and have no solution; if $M = N$, the system should be *exactly determined* with the solutions locally isolated; and if $M < N$, the system should be *under-determined* and the solutions not locally isolated. Clearly, all these statements are not always true (you can see this just by considering examples with linear equations). So, what does it mean to be in the “normal case”? The implicit function theorem provides an answer: one needs the equations (which we assume are differentiable) to be independent (that is, truly distinct) at the solutions. Definition 17.D.3 captures this notion.

Definition 17.D.3: The system of M equations in N unknowns $f(v) = 0$ is *regular* if $\text{rank } Df(v) = M$ whenever $f(v) = 0$.

For a regular system, the implicit function theorem (see Section M.E of the Mathematical Appendix) yields the existence of the right number of degrees of freedom. If $M < N$, we can choose M variables corresponding to M linearly independent columns of $Df(v)$ and we can express the values of these M variables that solve the M equations $f(v) = 0$ as a function of the $N - M$ remaining variables (see Exercise 17.D.2). If $M = N$, equilibria must be locally isolated for the same reasons as discussed earlier in this section for the system $\hat{z}(p) = 0$. And if $M > N$, then $\text{rank } Df(v) \leq N < M$ for all v ; in this case, Definition 17.D.3 simply says that, as a matter of definition, the equation system $f(v) = 0$ is regular if and only if the system admits no solution.

It remains to be argued that the regular case is the “normal” one. Figure 17.D.4 suggests how this can be approached. In the figure, the one-equation, one-unknown system $f(v) = 0$ is not regular [because of the tangency point of the graph of $f(\cdot)$ and the horizontal axis]. But clearly this phenomenon is not robust: if we slightly perturb the equation in an arbitrary manner [say that the shocked system is $f^s(\cdot)$], we get a regular system. On the other hand, the regularity of a system that is already regular is preserved for any small perturbation.¹⁸

This intuitive idea of a perturbation can be formalized as follows. Suppose there are some parameters $q = (q_1, \dots, q_S)$ such that, for every q , we have a system of equations $f(v; q) = 0$, as above. The set of possible parameter values is $\mathbb{R}^S$ (or an open region of $\mathbb{R}^S$). We can then justifiably say that $f(\cdot; q')$ is a perturbation of

18. The perturbation should control the values *and* the derivatives of the function. In technical language, it should be a C^1 perturbation.

$f(\cdot; q)$ if q' is close to q . Hence, the notion that the regularity of a system $f(\cdot; q) = 0$ is typical, or generic, could be captured by demanding that for almost every q , $f(\cdot; q) = 0$ be regular; in other words, that nonregular systems have probability zero of occurring (with respect to say, a nondegenerate normal distribution on $\mathbb{R}^S$).¹⁹ It stands to reason that some condition will be required on the dependence of $f(\cdot; q)$ on q for this to hold. At the very least, $f(\cdot; q)$ has to actually depend on q . The important mathematical theorem to be presented next tells us that little beyond this is needed.²⁰

Proposition 17.D.3: (The Transversality Theorem) If the $M \times (N + S)$ matrix $Df(v; q)$ has rank M whenever $f(v; q) = 0$ then for almost every q , the $M \times N$ matrix $D_v f(v; q)$ has rank M whenever $f(v; q) = 0$.

Heuristically, the assumption of the transversality theorem requires that there be enough variation in our universe. If $Df(v; q)$ has rank M whenever $f(v; q) = 0$, then from any solution it is always possible to (differentially) alter the values of the function f in any prescribed direction by adjusting the v and q variables. The conclusion of the theorem is that, if this can always be done, then whenever we are initially at a nonregular situation an arbitrary random displacement in q breaks us away from nonregularity. In fanciful language, if our universe is nondegenerate, then so will be almost every world in it. Note one of the strengths of the theorem: the matrix $Df(v; q)$ has M rows and $N + S$ columns. Hence, if S is large, so that there are many perturbation parameters, then the assumption of the theorem is likely to be satisfied; after all, we only need to find M linearly independent columns. On the other hand, $D_v f(v; q)$ has M rows but only N columns. It is thus harder to guarantee in advance that at a solution $D_v f(v; q)$ has M linearly independent columns. But the theorem tells us that this is so for almost every q . Observe that if $M > N$ (more equations than unknowns), then the $M \times N$ matrix $D_v f(v; q)$ cannot have rank M . Hence, the theorem tells us that in this case, generically (i.e., for almost every q), $f(v; q) = 0$ has no solution.

Let us now specialize our discussion to the case of a system of $L - 1$ excess demand equations in $L - 1$ unknowns, $\hat{z}(p) = 0$. We have seen by example that nonregular economies are possible. We wish to argue that they are not typical. To

19. More formally, we could say that in a system defined by finitely many parameters (taking values in, say, an open set) a property is *generic in the first sense* if it holds for a set of parameters of full measure (i.e., the complement of the set for which it holds has measure zero). The property is *generic in the second sense* if it holds in an open set of full measure. A full measure set is dense but it need not be open. Hence, the second sense is stronger than the first. Yet in many applications (all of ours in fact), the property under consideration holds in an open set, and so genericity in the first sense automatically yields genericity in the second sense. In some applications there is no finite number of parameters and no notion of measure to appeal to. In those cases we could say that a property is *generic in the third sense* if the property holds in an open and dense set. When no measure is available, this still provides a sensible way to capture the idea that the property is typical; but it should be noted that with finitely many parameters a set may be open, dense and have arbitrarily small (positive) measure. In this entire section we deal with genericity in the first sense, and we simply call it *genericity*.

20. For this theorem, we assume that $f(v; q)$ is as many times differentiable in its two arguments as is necessary.

do so, we could resort to a wide variety of perturbation parameters influencing preferences or endowments (or, in a more general setting, technologies). A natural set of parameters are the initial endowments themselves:

$$\omega = (\omega_{11}, \dots, \omega_{L1}, \dots, \omega_{1I}, \dots, \omega_{LI}) \in \mathbb{R}_{++}^{LI}.$$

We can write the dependence of the economy's excess demand function on endowments explicitly as $\hat{z}(p; \omega)$. We then have Proposition 17.D.4.

Proposition 17.D.4: For any p and ω , $\text{rank } D_{\omega} \hat{z}(p; \omega) = L - 1$.

Proof: It suffices to consider the endowments of a single consumer, say consumer 1, and to show that the $(L - 1) \times L$ matrix $D_{\omega_1} \hat{z}(p; \omega)$ has rank $L - 1$ [this implies that $\text{rank } D_{\omega} \hat{z}(p; \omega) = L - 1$]. To show this, we can either compute $D_{\omega_1} \hat{z}(p; \omega)$ explicitly (Exercise 17.D.3) or simply note that any perturbation of ω_1 , say $d\omega_1$, that leaves the wealth of consumer 1 at prices p unaltered will not change demand and therefore will change excess demand by exactly $-d\omega_1$. Specifically, if $p \cdot d\omega_1 = 0$ then, denoting $d\hat{\omega}_1 = (d\omega_{11}, \dots, d\omega_{L-1,1})$, we have $D_{\omega_1} \hat{z}(p; \omega) d\omega_1 = D_{\omega_1} \hat{z}_1(p; \omega) d\hat{\omega}_1 = -d\hat{\omega}_1$. Because the condition $p \cdot d\omega_1 = 0$ on $d\omega_1$ places no restrictions on $d\hat{\omega}_1$, it follows that by changing consumer 1's endowments we can move $\hat{z}(p; \omega)$ in any desired direction in $\mathbb{R}^{L-1}$, and so $\text{rank } D_{\omega_1} \hat{z}(p; \omega) = L - 1$. ■

We are now ready to state the main result [due to Debreu (1970)].

Proposition 17.D.5: For almost every vector of initial endowments $(\omega_1, \dots, \omega_I) \in \mathbb{R}_{++}^{LI}$, the economy defined by $\{(\succsim_i, \omega_i)\}_{i=1}^I$ is regular.²¹

Proof: Because of Proposition 17.D.4, the result follows from the transversality theorem (Proposition 17.D.3). ■

See Exercises 17.D.4 to 17.D.6 for variations on the theme of Proposition 17.D.4.

In Figure 17.D.5, we represent the equilibrium set $E = \{(\omega_1, \omega_2, p_1) : \hat{z}(p_1, 1; \omega) = 0\}$ of an Edgeworth box economy with total endowment $\bar{\omega} = \omega_1 + \omega_2$. The set E is the graph of the correspondence that assigns equilibrium prices to economies $\omega = (\omega_1, \omega_2)$.

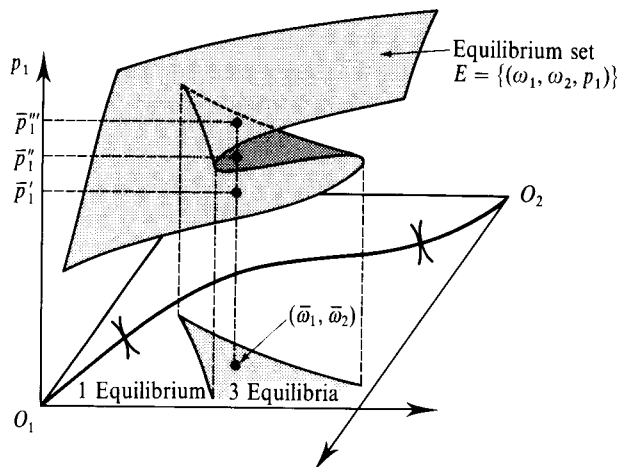


Figure 17.D.5
The equilibrium set

21. To be quite explicit, this means that the set of endowments that yield nonregular economies is a subset of $\mathbb{R}^{LI}$ that has $(LI\text{-dimensional})$ Lebesgue measure zero, or, equivalently, probability zero for, say, a nondegenerate $LI\text{-dimensional}$ normal distribution.

Because of the index theorem, this picture, in which the number of equilibria changes discontinuously from 3 to 1 at some points in the space of endowments is typical of the multiple-equilibrium case. A very extensive analysis of this equilibrium set has been carried out by Balasko (1988).

We conclude the discussion of genericity with two observations: First, the generic local determinateness of the theory extends to cases with externalities, taxes, or other “imperfections” leading to the failure of the first welfare theorem. (See Exercise 17.D.6.) This should be clear from the generality of the mathematical techniques which, in essence, rely only on the ability to express the equilibria of the theory as the zeros of a natural system of equations with the same number of equations and unknowns. Second, “finiteness of the number of equilibria” is a blunt conclusion. It is not the same if the “finite” stands for three or for a few million. Unfortunately, short of going all the way to uniqueness conditions (as we do in Section 17.F), we have no technique that allows us to refine our conclusions. We want to put on record, however, that it should not be presumed that in all generality “finite” means “small.” In this respect, we mention, tentatively, that there seems to be a distinction between market equilibrium situations for which the first welfare theorem holds (in which, indeed, examples with “many” equilibria seem contrived) and situations with a variety of market failures (where examples are easy to produce). See Exercise 17.D.7 and the discussion on “sunspots” in Section 19.F.

On the Index Theorem

The index result (Proposition 17.D.2) is, in its essence, a purely mathematical fact. An attempt at a rigorous proof would take us too far afield. Nonetheless, it is instructive to give an argument for its validity. It is an argument, we note incidentally, that can be made into a rigorous proof.

Denote our given, normalized, excess demand function by $\hat{z}(p)$. We begin by availing ourselves of some other excess demand function $\hat{z}^0(p)$ with the properties that (i) there is a unique $\bar{p}$ such that $\hat{z}^0(\bar{p}) = 0$ and (ii) $\text{sign}|D\hat{z}^0(\bar{p})| = (-1)^{L-1}$. For example, $\hat{z}^0(p)$ could be generated from a single-consumer Cobb–Douglas economy (Exercise 17.D.8). The idea is that $\hat{z}^0(p)$ is both simple and familiar to us and that, as a consequence, we can use it to learn about the properties of the unfamiliar $\hat{z}(p)$.

Consider the following one-parameter family (in technical language, a *homotopy*) of excess demand functions:

$$\hat{z}(p, t) = t\hat{z}(p) + (1 - t)\hat{z}^0(p) \quad \text{for } 0 \leq t \leq 1.$$

The system $\hat{z}(p, t) = 0$ has $L - 1$ equations and L unknowns: $(p_1, \dots, p_{L-1}, t)$. Typically, therefore, the solution set $E = \{(p, t) : \hat{z}(p, t) = 0\}$ has *one and only one* degree of freedom at any of its points (that is, it looks locally like a segment). Moreover, since this solution set cannot escape to infinite or zero prices (because of the boundary conditions on excess demand) and is closed [because of the continuity of $\hat{z}(p, t)$], it follows that the general situation is well represented in Figure 17.D.6.

In Figure 17.D.6, we depict E as formed, so to speak, by a finite number of circle-like and segment-like components, with the endpoints of the segments at the $t = 0$ and $t = 1$ boundaries. Since there are two endpoints per segment, there is an even number of such endpoints. By construction, $\bar{p}$ is the only endpoint at the $t = 0$ boundary.²² Therefore, there must be an odd number of endpoints at the $t = 1$ boundary; that is, there is an odd number of solutions to $\hat{z}(p) = \hat{z}(p, 1) = 0$. Suppose now that we follow a segment from end to end. What

22. More generally, if $\hat{z}(p; t)$ is an arbitrary homotopy then the typical situation is well represented by any of the Figures 17.G.1(a), (b), or (c).

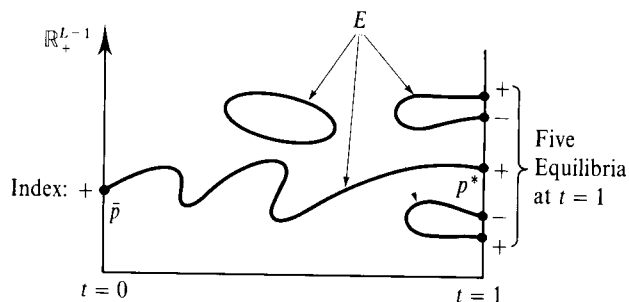


Figure 17.D.6
The equilibrium set
under a homotopy.

is the relation between the indices at the two ends? A moment's reflection (keeping the implicit function theorem in mind) reveals that as long as we move in a given direction relative to t (i.e., forward or backward), the index, $(-1)^{L-1} \text{sign } |D_p \hat{z}(p, t)|$, does not change, and that the index changes sign precisely when we reverse direction.²³ Now, a segment that begins and ends at the same boundary must reverse direction an odd number of times; hence, the indices at the two endpoints have opposite signs. You can verify this in Figure 17.D.6. Therefore, the sum of the indices at $t = 1$ equals the index of the lone equilibrium of $\hat{z}(\cdot)$ connected by a segment to the equilibrium $\bar{p}$ of $\hat{z}^0(\cdot)$ at the boundary $t = 0$. It is represented by p^* in Figure 17.D.6. The segment that connects $\bar{p}$ to p^* in E reverses directions an even number of times (possibly none); therefore, we conclude that the index of this equilibrium at $t = 1$ equals the index of $\bar{p}$ for $\hat{z}^0(\cdot)$, which, by construction, is $+1$. Hence, the sum of the indices at $t = 1$ is $+1$, as Proposition 17.D.2 asserts to be true in complete generality.

17.E Anything Goes: The Sonnenschein–Mantel–Debreu Theorem

We have seen that under a number of general assumptions (of which the most substantial concerns convexity), an equilibrium exists and the number of equilibria is typically finite. Those are important properties, but we would like to know if we could say more, especially for predictive or comparative-statics purposes (see Section 17.G). We may well suspect by now (especially if the message of Chapter 4 on the difficulties of demand aggregation has been well understood) that the answer is likely to be negative; that is, that, in general, we will not be able to impose further restrictions on excess demand than those in Proposition 17.B.2, and therefore that no further general restrictions on the nature of Walrasian equilibria than those already studied can be hoped for. Special assumptions will have to be made to derive stronger implications (such as uniqueness; see Section 17.F).

In this section, we confirm this and bring home the negative message in a particularly strong manner. The theme, culminating in Propositions 17.E.3 and 17.E.4, is: *Anything satisfying the few properties that we have already shown must hold, can actually occur.*

23. To see this, think of the case where $L = 2$. Applying the implicit function theorem to $\hat{z}_1(p_1, t) = 0$, verify then that a reversal of direction occurs precisely where $\partial \hat{z}_1(p_1, t) / \partial p_1 = 0$.

The analysis that follows develops the logic of this conclusion through a series of intermediate results that have independent interest. Some readers may wish, in a first reading of this section, to skip these results and examine directly the statements of Propositions 17.E.3 and 17.E.4 and the accompanying discussion of their interpretations.

To be specific, we concentrate the analysis, as usual, on exchange economies formalized by means of excess demand equations. Focusing on exchange economies makes sense because, as we know from Chapter 5, aggregation effects are unproblematic in production. The source of the aggregation problem rests squarely with the wealth effects of the consumption side.

We begin by posing a relatively simple but nonetheless quite important question: To what extent can we derive restrictions on the behavior of excess demand at a given price p . In particular, we ask for possible restrictions on the $L \times L$ matrix of price effects $Dz(p)$.²⁴

Suppose that $z(p)$ is a differentiable aggregate excess demand function. In Exercise 17.E.1, you are asked to show that

$$\sum_k \frac{\partial z_\ell(p)}{\partial p_k} p_k = 0 \quad \text{for all } \ell \text{ and } p \text{ [or } Dz(p)p = 0] \quad (17.E.1)$$

$$\sum_k p_k \frac{\partial z_k(p)}{\partial p_\ell} = -z_\ell(p) \quad \text{for all } \ell \text{ and } p \text{ [or } p \cdot Dz(p) = -z(p)] \quad (17.E.2)$$

These are the excess demand counterparts of expressions (2.E.1) and (2.E.4) for demand functions. They follow, respectively, from the homogeneity of degree zero and the Walras' law properties of excess demand. More interestingly, from $z(p) = \sum_i (x_i(p, p \cdot \omega_i) - \omega_i)$ we also get

$$Dz(p) = \sum_i [S_i(p, p \cdot \omega_i) - D_{w_i} x_i(p, p \cdot \omega_i) z_i(p)^T] \quad (17.E.3)$$

where, as usual, $S_i(p, p \cdot \omega_i)$ is the substitution matrix (see Exercise 17.E.2).

Expression (17.E.3) is very instructive. It tells us that if it were not for the wealth effects, $Dz(p)$ would inherit the negative semidefiniteness (n.s.d.) property of the substitution matrices. How much havoc can the wealth effects cause? Notice that the matrix

$$D_{w_i} x_i(p, p \cdot \omega_i) z_i(p)^T = \begin{bmatrix} \frac{\partial x_{1i}(p, p \cdot \omega_i)}{\partial w_i} z_{1i}(p) & \dots & \frac{\partial x_{1i}(p, p \cdot \omega_i)}{\partial w_i} z_{Li}(p) \\ & \ddots & \\ \frac{\partial x_{Li}(p, p \cdot \omega_i)}{\partial w_i} z_{1i}(p) & \dots & \frac{\partial x_{Li}(p, p \cdot \omega_i)}{\partial w_i} z_{Li}(p) \end{bmatrix}$$

is of rank 1 (any two columns, or rows, are proportional). Therefore, we could informally surmise that the wealth effect of consumer i can hurt in at most one

24. Note that $z(p)$ can take any value. You need only specify a consumer with an endowment vector ω such that $\omega + z(p) \gg 0$ and then choose a utility function that has $\omega + z(p)$ as the demanded point.

direction of price change.²⁵ Thus, we should expect that if $I < L$ then there are some negative semidefiniteness restrictions left on $Dz(p)$. That this is the case is formalized in Proposition 17.E.1.

Proposition 17.E.1: Suppose that $I < L$. Then for any equilibrium price vector p there is some direction of price change $dp \neq 0$ such that $p \cdot dp = 0$ (hence, dp is not proportional to p) and $dp \cdot Dz(p) dp \leq 0$.

Proof: Because $z(p) = \sum_i z_i(p) = 0$, at most I of the $I + 1$ vectors,

$$\{p, z_1(p), \dots, z_I(p)\} \subset \mathbb{R}^L$$

can be linearly independent. Since $I < L$, it follows that we can find a nonzero vector $dp \in \mathbb{R}^L$ such that $p \cdot dp = 0$ and $z_i(p) \cdot dp = 0$ for all i . In words: dp is a nonproportional price change that is compensated (i.e., there is no change in real wealth) for every consumer. But then from (17.E.3) we obtain

$$dp \cdot Dz(p) dp = \sum_i dp \cdot S_i(p, p \cdot \omega_i) dp \leq 0. \quad \blacksquare$$

Parallel reasoning should make us expect that if $I \geq L$ (i.e., if there are at least as many consumers as commodities), then there may not be any restriction left on $Dz(p)$ beyond (17.E.1) and (17.E.2). After all, the direction of an individual wealth effect vector at a given price is quite arbitrary (and can be chosen independently of the substitution effects of the corresponding individual); and with $I \geq L$ wealth effect vectors to be specified, there is considerable room to maneuver. Proposition 17.E.2 confirms this suspicion.

Proposition 17.E.2: Given a price vector p , let $z \in \mathbb{R}^L$ be an arbitrary vector and A an arbitrary $L \times L$ matrix satisfying $p \cdot z = 0$, $Ap = 0$ and $p \cdot A = -z$. Then there is a collection of L consumers generating an aggregate excess demand function $z(\cdot)$ such that $z(p) = z$ and $Dz(p) = A$.

Proof: To keep the argument simple, we restrict ourselves to a search for consumers that at their demanded vectors have a null substitution matrix, $S_i(p, p \cdot \omega_i) = 0$, that is, whose indifference sets exhibit a vertex at the chosen point.²⁶

We can always formally rewrite the given $L \times L$ matrix A as

$$A = \sum_{\ell} e^{\ell} a^{\ell},$$

where e^{ℓ} is the ℓ th unit column vector (i.e., all the entries of e^{ℓ} are 0 except the ℓ th entry, which equals 1) and a^{ℓ} is the ℓ th row of A [i.e., $a^{\ell} = (a_{\ell 1}, \dots, a_{\ell L})$].

Suppose now that we could specify L consumers, $i = 1, \dots, L$, with the property that, for every i , consumer i has, at the price vector p , an excess demand vector $z_i(p) = -p_i(a^i)^T$, a wealth effect vector $D_{w_i}x_i(p, p \cdot \omega_i) = (1/p_i)e^i$, and a substitution matrix $S_i(p, p \cdot \omega_i) = 0$ (where $a^1, \dots, a^L$ and $e^1, \dots, e^L$ are as defined above). Then we would have both

$$z(p) = \sum_i z_i(p) = -\sum_{\ell} p_{\ell}(a^{\ell})^T = -A^T p = -p \cdot A = z$$

25. For example, it cannot hurt in any direction of price change that is orthogonal to the wealth effects vector $D_{w_i}x_i(p, p \cdot \omega_i)$ or to the excess demand vector $z_i(p)$. A more precise argument is given in Proposition 17.E.1.

26. The term "vertex" refers to what is usually called a "kink" in the case $L = 2$.

and

$$\begin{aligned} Dz(p) &= -\sum_i D_{\omega_i} x_i(p, p \cdot \omega_i) z_i(p)^T \\ &= \sum_\ell (1/p_\ell) e^\ell (p_\ell a^\ell) = \sum_\ell e^\ell a^\ell = A, \end{aligned}$$

and so we would have accomplished our objective.

Can we find these L consumers? The answer is “yes.” Begin by choosing a collection of endowments $(\omega_1, \dots, \omega_L)$ yielding strictly positive consumptions when excess demands are $z_i(p) = -p_i(a^i)^T$; that is, $x_i = \omega_i - p_i(a^i)^T \gg 0$ for every i . Observe then that, for every $i = 1, \dots, L$, the candidate individual excess demand satisfies Walras’ law

$$p \cdot z_i(p) = -p_i p \cdot a^i = 0 \quad (\text{because } Ap = 0),$$

and, also, that the candidate wealth effect vector satisfies the necessary condition of Proposition 2.E.3

$$p \cdot D_{\omega_i} x_i(p, p \cdot \omega_i) = (1/p_i) p \cdot e^i = 1.$$

Figure 17.E.1 should then be persuasive enough in convincing us that we can assign preferences to $i = 1, \dots, L$ in such a way that the chosen consumption at p is x_i , the wealth effect vector at p is proportional to e^i (and therefore must equal $(1/p_i)e^i$),²⁷ and the indifference map has a kink at x_i . The figure illustrates the complete construction for the case $L = 2$.²⁸ In Exercise 17.E.3, you are asked to write an explicit utility function. ■

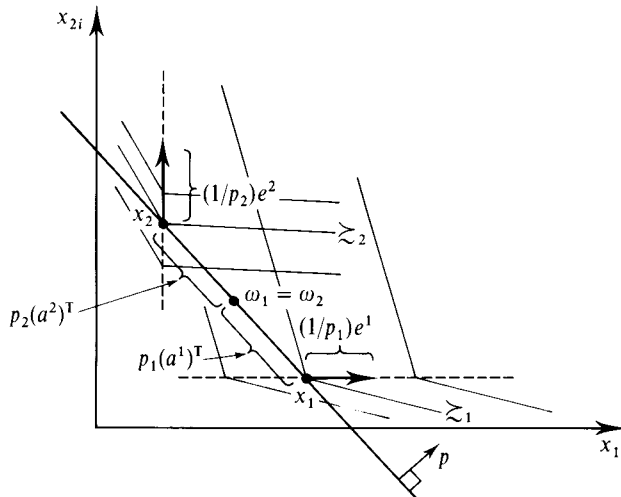


Figure 17.E.1

Decomposition of excess demand and price effects at a price vector p (for $L = 2$).

27. Indeed, if $Dx_i(p, p \cdot \omega_i) = \alpha_i e^i$, then $1 = p \cdot Dx_i(p, p \cdot \omega_i) = \alpha_i p \cdot e^i = \alpha_i p_i$. Hence, $\alpha_i = 1/p_i$.

28. At no extra cost, we could actually accomplish a bit more. We could also require the substitution matrices of the consumers $i = 1, \dots, L$ to be any arbitrary collection of $L \times L$ matrices S_i satisfying the properties: S_i is symmetric, negative semidefinite, $p \cdot S_i = 0$, and $S_i p = 0$. The specification of consumers generating excess demand $z(p)$ and excess demand effects $Dz(p)$ at p would proceed in a manner similar to the proof just given except that the argument would now be applied to $A - \sum_i S_i$. By using matrices S_i of maximal rank (i.e., of rank $L - 1$), we could insure that the resulting L consumers display smooth indifference sets at their chosen consumptions.

Up to now, we have studied the possibility of restrictions on the behavior of excess demand at a single price vector. Although the results of Propositions 17.E.1 and 17.E.2 are already quite useful, we can go further. The essence of the negative point being made is, unfortunately, much more general. Consider an arbitrary function $z(p)$, and let us for the moment sidestep boundary issues by having $z(p)$ be defined on a domain where relative prices are bounded away from zero; that is, for a small constant $\varepsilon > 0$, we consider only price vectors p with $p_\ell/p_{\ell'} \geq \varepsilon$ for every ℓ and ℓ' . We could then ask: "Can $z(\cdot)$ coincide with the excess demand function of an economy for every p in its domain?" Of course, in its domain, $z(\cdot)$ must fulfill three obvious necessary conditions: it must be continuous, it must be homogeneous of degree zero, and it must satisfy Walras' law. But for any $z(\cdot)$ satisfying these three conditions, it turns out that the answer is, again, "yes."²⁹

Proposition 17.E.3: Suppose that $z(\cdot)$ is a continuous function defined on

$$P_\varepsilon = \{p \in \mathbb{R}_+^L : p_\ell/p_{\ell'} \geq \varepsilon \text{ for every } \ell \text{ and } \ell'\}$$

and with values in $\mathbb{R}^L$. Assume that, in addition, $z(\cdot)$ is homogeneous of degree zero and satisfies Walras' law. Then there is an economy of L consumers whose aggregate excess demand function coincides with $z(p)$ in the domain P_ε .³⁰

Proof: At the end of this section, we offer (in small-type) a brief discussion of the general proof of this result. Here, we limit ourselves to the comparatively simple case where $L = 2$.

Suppose then that $L = 2$ and that an $\varepsilon > 0$ and a function $z(\cdot)$ satisfying the assumption of the proposition are given to us. The continuity and homogeneity of degree zero of $z(\cdot)$ imply the existence of a number $r > 0$ such that $|z_1(p)| < r$ for every $p \in P_\varepsilon$. We now specify two functions $z^1(\cdot)$ and $z^2(\cdot)$ with domain P_ε and values in $\mathbb{R}^2$, which are also continuous and homogeneous of degree zero, and satisfy Walras' law. In particular, we let

$$z_1^1(p) = \frac{1}{2}z_1(p) + r \quad [\text{accordingly, } z_2^1(p) = -(p_1/p_2)z_1^1(p)]$$

and

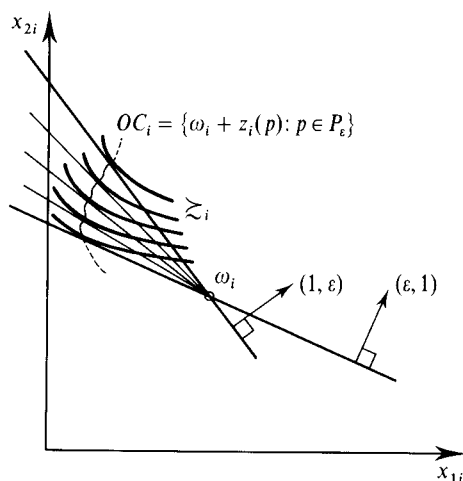
$$z_1^2(p) = \frac{1}{2}z_1(p) - r \quad [\text{accordingly, } z_2^2(p) = -(p_1/p_2)z_1^2(p)].$$

Note that $z(p) = z^1(p) + z^2(p)$ for every $p \in P_\varepsilon$. We shall show that for $i = 1, 2$ the function $z^i(\cdot)$ coincides in the domain P_ε with the excess demand function of a consumer. To this effect, we use the following properties of $z^i(\cdot)$: continuity, homogeneity of degree zero, satisfaction of Walras law, and the fact that there is no $p \in P_\varepsilon$ such that $z^i(p) = 0$. In Exercise 17.E.4, you are asked to show by example that this last requirement is needed.

Choose a $\omega_i \gg 0$ such that $\omega_i + z^i(p) \gg 0$ for every $p \in P_\varepsilon$. In Figure 17.E.2, we represent the offer curve OC_i associated with $z^i(\cdot)$ in the domain P_ε . In the figure, for every $p \in P_\varepsilon$,

29. The question was posed by Sonnenschein (1973). He conjectured that the answer was that, indeed, on the domain where $p_\ell \geq \varepsilon$ for all ℓ , the three properties were not only necessary but also sufficient; that is, we could always find such an economy. He also proved that this is so for the two-commodity case. The problem was then solved by Mantel (1974) for any number of commodities. Mantel made use of $2L$ consumers. Shortly afterwards, Debreu (1974) gave a different and very simple proof requiring the indispensable minimum of L consumers. This was topped by Mantel (1976), who refined his earlier proof to show that L homothetic consumers (with no restrictions in their initial endowments) would do.

30. Note, in particular, that this result implies that for any $I \geq L$, there is an economy of I consumers that generates $z(\cdot)$ on P_ε . We need only add to the L consumers identified by the proposition $I - L$ consumers who have no endowments (or, alternatively, whose most preferred consumption bundle at all price vectors in P_ε is their endowment vector).

**Figure 17.E.2**

Construction of preferences (in the case $L = 2$) for the offer curve of an excess demand function $z_i(\cdot)$ such that $z_i(p) = 0$ has no solution with $1/\epsilon < p_1/p_2 < \epsilon$.

$\omega_i + z^i(p)$ is the intersection point of the offer curve with the budget line perpendicular to p . The offer curve is continuous and, because $z^i(p) = 0$ has no solution in P_ϵ , it does not touch the initial endowment point. We then see in the figure that no matter how complicated the offer curve may otherwise be, we can always fit an indifference map so that for any $p \in P_\epsilon$ we generate precisely the demands $\omega_i + z^i(p)$. ■

Strictly speaking, Proposition 17.E.3 does not yet settle our original question, “Can we assert anything more about the equilibria of an economy than what we have derived in Sections 17.C and 17.D?” The problem is that Proposition 17.E.3 characterizes the behavior of excess demand away from the boundary, whereas it is the power of the boundary conditions that yields some of the restrictions we have already established: existence, (generic) finiteness, oddness, the index formula.³¹ To argue that we cannot hope for more restrictions than these on the equilibrium set, we need to guarantee that if a candidate equilibrium set satisfies them, then the construction of the “explaining” economy will not add new equilibria. The result presented in Proposition 17.E.4, whose proof we omit, provides therefore the final answer to our question.³²

Proposition 17.E.4: For any $N \geq 1$, suppose that we assign to each $n = 1, \dots, N$ a price vector p^n , normalized to $\|p^n\| = 1$, and an $L \times L$ matrix A_n of rank $L - 1$, satisfying $A_n p^n = 0$ and $p^n \cdot A_n = 0$. Suppose that, in addition, the index formula $\sum_n (-1)^{L-1} \text{sign } |\hat{A}_n| = +1$ holds.³³ If $L = 2$, assume also that positive and negative index equilibria alternate.

Then there is an economy with L consumers such that the aggregate excess demand $z(\cdot)$ has the properties:

- (i) $z(p) = 0$ for $\|p\| = 1$ if and only if $p = p^n$ for some n .
- (ii) $Dz(p^n) = A_n$ for every n .

31. Note, for example, that although a candidate function $z(\cdot)$ defined on P_ϵ may not have any solution, we can still successfully generate it from an economy. What happens, of course, is that the equilibria of the economy (which must exist) are all outside of P_ϵ .

32. For this and more general results, see Mas-Colell (1977).

33. Here, $\hat{A}_n$ is the $L - 1 \times L - 1$ matrix obtained by deleting one row and corresponding column from A .

Proposition 17.E.4 tells us that for any finite collection of price vectors $\{p^1, \dots, p^N\}$ and matrices of price effects $\{Dz(p^1), \dots, Dz(p^N)\}$, we can find an economy with L consumers for which these price vectors are equilibrium price vectors and $\{Dz(p^1), \dots, Dz(p^N)\}$, are the corresponding price effects at these equilibria. The result implies that to derive further restrictions on Walrasian equilibria we will need to make additional (and, as we shall see, strong) assumptions. This is the subject of the next three sections. An excellent survey for further reading on the topic of this section is Shafer and Sonnenschein (1982).

We should point out that the initial endowments of the consumers obtained by means of Propositions 17.E.2, 17.E.3 or 17.E.4 are not a priori limited in any way. If there are constraints on permissible initial endowments, the nonnegativity conditions on consumption come into play and there may, in fact, be other restrictions on the function $z(\cdot)$. For example, you are asked in Exercise 17.E.5 to verify that the excess demand vectors $z(p)$ and $z(p')$ represented in Figure 17.E.3 cannot be decomposed into individual excess demand functions generated by rational preferences if the amount of any commodity that any consumer may possess as an initial endowment is prescribed to be at most 1 and if consumptions must be nonnegative.

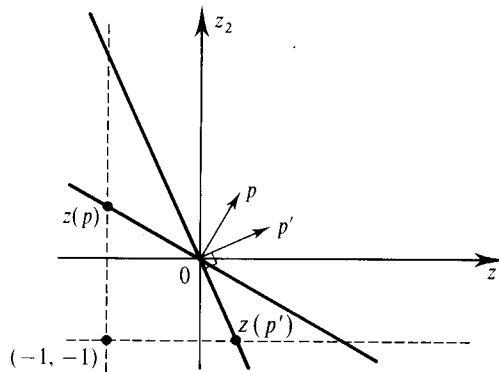


Figure 17.E.3
Excess demand choices
that cannot be
decomposed due to
boundary constraints.

Proof of Proposition 17.E.3 continued: Although a complete proof of the proposition for the case of any number of commodities would take us too far afield, the essentials of the proof by Debreu (1974) are actually not too difficult to convey. We shall attempt to do so. We note that, when carefully examined, the proof can be seen as a generalization of the argument for the $L = 2$ case presented earlier.

In Section 3.J, we saw that the strong axiom of revealed preference (SA) for demand functions is equivalent to the existence of rationalizing preferences. The same is true for excess demand functions: If an excess demand function $z^i(\cdot)$ satisfies the SA (we will give a precise definition in a moment), then $z^i(\cdot)$ can be generated from rational preferences.³⁴ It is thus reasonable to redefine our problem as: Given a function $z(\cdot)$ that, on the domain P_e , is continuous, homogeneous of degree zero, and satisfies Walras' law (for short, we refer to these functions as *excess demand functions*), can we find L excess demand functions $z^i(\cdot)$, each satisfying the SA, such that $\sum_i z^i(p) = z(p)$ for every $p \in P_e$?

Before proceeding, let us define the SA for an excess demand function $z^i(\cdot)$. The definition is just a natural adaptation of the definition for demand functions. We say that p is *directly revealed preferred* to p' if

34. We refer to the proof of Proposition 3.J.1 for the justification of this claim.

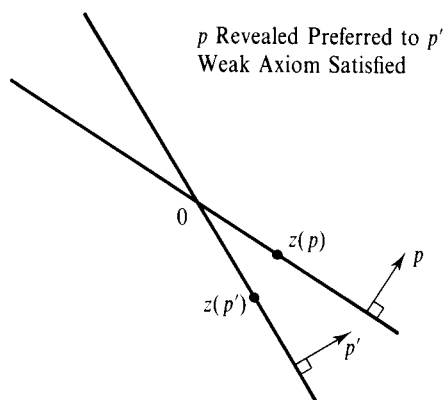


Figure 17.E.4
Revealed preference
for excess demand.

$$z^i(p) \neq z^i(p') \quad \text{and} \quad p \cdot z^i(p') \leq 0 \quad (17.E.4)$$

(see Figure 17.E.4). We say that p is *indirectly revealed preferred* to p' if there is a finite chain $p^1, \dots, p^N$ such that $p^1 = p$, $p^N = p'$, and p^n is directly revealed preferred to p^{n+1} for all $n \leq N - 1$. The SA then says:

For every p and p' , if p is (directly or indirectly) revealed preferred to p' , then p' cannot be (directly) revealed preferred to p .

From now on, we let prices be normalized. A convenient normalization here is $\|p\|^2 = p \cdot p = \sum_i (p_i)^2 = 1$.

We say that an excess demand function $z^i(\cdot)$ is *proportionally one-to-one* if $p \neq p'$ implies that $z^i(p)$ is not proportional to $z^i(p')$; in particular, we have $z^i(p) \neq z^i(p')$. For proportionally one-to-one excess demand functions (and normalized prices), we can restate the “directly revealed preferred” definition (17.E.4) as

$$p \neq p' \quad \text{and} \quad p \cdot z^i(p') \leq 0. \quad (17.E.4')$$

Suppose that $\alpha_i(\cdot)$ is an arbitrary real-valued function of p such that $\alpha_i(p) > 0$ for all $p \in P_e$. The basic observation of the proof is then the following: if $z^i(\cdot)$ is a *proportionally one-to-one* excess demand function that satisfies the SA, then the same properties are true of the function $\alpha_i(\cdot)z^i(\cdot)$. Indeed, for any p and p' the revealed preference inequalities (17.E.4') hold for $z^i(\cdot)$ if and only if they hold for $\alpha_i(\cdot)z^i(\cdot)$, and if $z^i(\cdot)$ is proportionally one-to-one, then so is $\alpha_i(\cdot)z^i(\cdot)$. This observation suggests a way to proceed. We could look for L proportionally one-to-one excess demand functions $z^i(\cdot)$ satisfying the SA and such that, at every $p \in P_e$, the vectors $\{z^1(p), \dots, z^L(p)\}$ constitute a basis capable of spanning $z(p)$ by means of a strictly positive linear combination, that is, such that for every $p \in P_e$ we can write $z(p) = \sum_i \alpha_i(p)z^i(p)$ for some numbers $\alpha_i(p) > 0$. This is precisely what we will now do.

For every normalized $p \in P_e$, denote $T_p = \{z \in \mathbb{R}^L : p \cdot z = 0\}$ and for every $i = 1, \dots, L$, let $z^i(p) \in T_p$ be the point that minimizes the Euclidean distance $\|z - e^i\|$ (or, equivalently, maximizes the concave “utility function” $-\|z - e^i\|$) for $z \in T_p$, where e^i is the i th unit vector (the column vector whose i th entry is 1 with zeros elsewhere). Geometrically, $z^i(p)$ is the perpendicular projection of e^i on the budget hyperplane T_p ; that is, $z^i(p) = e^i - p_i p$, where p_i is the i th component of the vector p (recall that $i \leq L$). Then $z^i(\cdot)$ is proportionally one-to-one (see Exercise 17.E.6) and satisfies the SA (since it is derived from utility maximization; see also Exercise 17.E.7).

Now let $r > 0$ be a large-enough number for us to have $z(p) + rp \gg 0$ for every normalized $p \in P_e$ [such an r exists by the continuity of $z(\cdot)$ and the fact that the set of normalized price vectors in P_e is compact and includes only strictly positive price vectors]. For every $i = 1, \dots, L$ and every normalized $p \in P_e$, define $\alpha_i(p) = z_i(p) + rp_i > 0$, where $z_i(p)$ is the i th component

The Weak Axiom for Aggregate Excess Demand

Suppose that the production side of the economy is given to us by an arbitrary technology $Y \subset \mathbb{R}^L$ of the constant returns, convex type (i.e., Y is a convex cone). What conditions involving only the demand side of the economy guarantee the uniqueness of equilibrium allocations?³⁶ From the analysis presented in Section 4.D, we already know one answer: if a welfare authority makes sure that wealth is always distributed so as to maximize a (strictly concave) social welfare function, then the economy admits a (strictly concave) normative representative consumer, and the equilibrium necessarily corresponds to the unique Pareto optimum of this one-consumer economy (as in Section 15.C). In our current framework, however, this is not a promising approach because wealth is derived from initial endowments and only by coincidence can we expect that the induced distribution of wealth maximizes a social welfare function. We will therefore concentrate on a weaker and, for the purpose at hand, more interesting condition: that the *weak axiom of revealed preference* holds for the aggregate excess demand of the consumers.

To begin, suppose that $z(p) = \sum_i (x_i(p, p \cdot \omega_i) - \omega_i)$ is the aggregate excess demand function of the consumers. For this economy with production, Proposition 17.F.1 provides a useful restatement of the definition of Walrasian equilibrium (Definition 17.B.1) in terms of $z(\cdot)$.

Proposition 17.F.1: Given an economy specified by the constant returns technology Y and the aggregate excess demand function of the consumers $z(\cdot)$, a price vector p is a Walrasian equilibrium price vector if and only if

- (i) $p \cdot y \leq 0$ for every $y \in Y$, and
- (ii) $z(p)$ is a feasible production; that is, $z(p) \in Y$.

Proof: If p is a Walrasian equilibrium price vector, then (ii) follows from market clearing and (i) is a necessary condition for profit maximization with a constant returns technology. In the other direction, if (i) and (ii) hold, then consumptions $x_i^* = x_i(p, p \cdot \omega_i)$ for $i = 1, \dots, I$, production vector $y^* = z(p) \in Y$, and price vector p constitute a Walrasian equilibrium. To verify this, the only condition that is not immediate is profit maximization. However, $y^* = z(p) \in Y$ is profit maximizing because $p \cdot y \leq 0$ for all $y \in Y$ [since (i) holds] and $p \cdot y^* = p \cdot z(p) = 0$ (from Walras' law). ■

We next define the weak axiom for excess demand functions.

Definition 17.F.1: (*The Weak Axiom for Excess Demand Functions*) The excess demand function $z(\cdot)$ satisfies the weak axiom of revealed preference (WA) if for any pair of price vectors p and p' , we have

$$z(p) \neq z(p') \text{ and } p \cdot z(p') \leq 0 \text{ implies } p' \cdot z(p) > 0.$$

36. The set Y can be thought as an aggregate production set. The restriction that Y be of constant returns is made merely for convenience of exposition. It allows us, for example, not to worry about the distribution of profits to consumers (since profits are zero in any equilibrium). Note also that the constant returns model includes pure exchange as a special case (where $Y = -\mathbb{R}_+^L$).

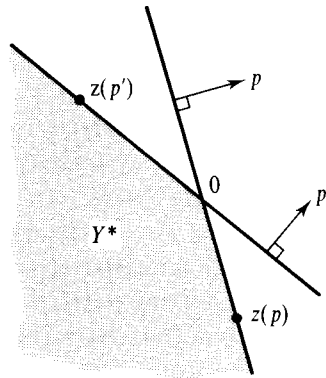


Figure 17.F.1
A violation of the weak axiom implies multiplicity of equilibria for some Y .

In words, the definition says that if p is revealed preferred to p' , then p' cannot be revealed preferred to p [i.e., $z(p)$ cannot be affordable under p']. It is the same definition used in Sections 1.G and 2.F, but now applied to excess demand functions.³⁷ The axiom is always satisfied by the excess demand function of a single individual, but it is a strong condition for *aggregate* excess demand (see Section 4.C for a discussion of this point).

We first note that, given $z(\cdot)$, the WA is a necessary condition for us to be assured of a unique equilibrium for every possible convex, constant returns technology Y that $z(\cdot)$ is coupled with. To see this, suppose that the WA was violated; that is, suppose that for some p and p' we have $z(p) \neq z(p')$, $p \cdot z(p') \leq 0$, and $p' \cdot z(p) \leq 0$. Then we claim that both p and p' are equilibrium prices for the convex, constant returns production set given by

$$Y^* = \{y \in \mathbb{R}^L : p \cdot y \leq 0 \text{ and } p' \cdot y \leq 0\}.$$

Figure 17.F.1 depicts this production set for the case $L = 2$. Note that we have $z(p) \in Y^*$ and $p \cdot y \leq 0$ for every $y \in Y^*$. Thus, by Proposition 17.F.1, p is an equilibrium price vector. The same is true for p' . Since $z(p) \neq z(p')$, we conclude that the equilibrium is not unique for the economy formed by $z(\cdot)$ and the production set Y^* .

What about sufficiency? The weak axiom is not quite a sufficient condition for uniqueness, but Proposition 17.F.2 shows that it does guarantee that for any convex, constant returns Y , the set of equilibrium price vectors is convex. Although this convexity property is certainly not the same as uniqueness, it has an immediate uniqueness implication: if an economy has only a *finite* number of (normalized) price equilibria (a generic situation according to Section 17.D),³⁸ the equilibrium must be unique.

37. A formal, and inessential, difference is that we now define the revealed preference relation on the budget sets (i.e., on price vectors) directly rather than on the choices (i.e., on commodity vectors).

38. Although our discussion in Section 17.D focused on the case of exchange economies, its conclusions regarding generic local uniqueness and finiteness of the equilibrium set can be extended to the present production context.

Proposition 17.F.2: Suppose that the excess demand function $z(\cdot)$ is such that, for any constant returns convex technology Y , the economy formed by $z(\cdot)$ and Y has a unique (normalized) equilibrium price vector. Then $z(\cdot)$ satisfies the weak axiom. Conversely, if $z(\cdot)$ satisfies the weak axiom then, for any constant returns convex technology Y , the set of equilibrium price vectors is convex (and so, if the set of normalized price equilibria is finite, there can be at most one normalized price equilibrium).

Proof: The first part has already been shown. To verify the convexity of the set of equilibrium prices, suppose that p and p' are equilibrium price vectors for the constant returns convex technology Y ; that is, $z(p) \in Y$, $z(p') \in Y$, and, for any $y \in Y$, $p \cdot y \leq 0$ and $p' \cdot y \leq 0$. Let $p'' = \alpha p + (1 - \alpha)p'$ for $\alpha \in [0, 1]$. Note, first, that $p'' \cdot y = \alpha p \cdot y + (1 - \alpha)p' \cdot y \leq 0$ for any $y \in Y$. To show that p'' is an equilibrium, we therefore need only establish that $z(p'') \in Y$. Because $0 = p'' \cdot z(p'') = \alpha p \cdot z(p'') + (1 - \alpha)p' \cdot z(p'')$, we have that either $p \cdot z(p'') \leq 0$ or $p' \cdot z(p'') \leq 0$. Suppose that the first possibility holds, so that $p \cdot z(p'') \leq 0$ [a parallel argument applies if, instead, $p' \cdot z(p'') \leq 0$]. Since $z(p) \in Y$ we have $p'' \cdot z(p) \leq 0$. But with $p'' \cdot z(p) \leq 0$ and $p \cdot z(p'') \leq 0$, a contradiction to the WA can be avoided only if $z(p'') = z(p)$. Hence $z(p'') \in Y$.³⁹ ■

We are therefore led to focus attention on conditions on preferences and endowments of the I consumers guaranteeing that the aggregate excess demand function $z(p)$ fulfills the WA. To begin with a relatively simple case, suppose that all the endowment vectors ω_i are proportional among themselves; that is, that $\omega_i = \alpha_i \bar{\omega}$, where $\bar{\omega}$ is the vector of total endowments and $\alpha_i \geq 0$ are shares with $\sum_i \alpha_i = 1$. In such an economy, the distribution of wealth across consumers is independent of prices. Normalizing prices to $p \cdot \bar{\omega} = 1$, the wealth of consumer 1 is α_i and $z_i(p) = x_i(p, \alpha_i) - \omega_i$. The aggregate demand behavior of a population of consumers with fixed wealth levels was studied in Section 4.C. We repeat our qualitative conclusion from there: if individual wealth levels remain fixed, the satisfaction of the WA by aggregate demand (or excess demand), although restrictive, is not implausible.⁴⁰

A proportionality assumption on initial endowments is not very tenable in a general equilibrium context. It is important, therefore, to ask which new effects are at work (relative to those studied in Section 4.C) when the distribution of endowments does not satisfy this hypothesis. Unfortunately, it turns out that nonproportionality of endowments can reduce the likelihood of satisfaction of the weak axiom by aggregate excess demand. To see this, consider the relatively simple situation in which preferences are homothetic. Recall from Sections 4.C and 4.D that, when endowments are proportional, this case is extremely well behaved; not only is the WA satisfied, but the model even admits a representative consumer. Yet, as we proceed to discuss

39. Observe that we have established that either $z(p'') = z(p)$ or $z(p'') = z(p')$. Since this is true for any $\alpha \in [0, 1]$, and since the function $z(\cdot)$ is continuous, this implies that $z(p) = z(p')$ for any two equilibrium price vectors p and p' ; that is, if the WA holds for $z(\cdot)$, then every Walrasian equilibrium for the given endowments must have the same aggregate consumption vector and, hence, the same aggregate production vector.

40. On this point, consult also the references given in Chapter 4, especially Hildenbrand (1994).

below (in small type), even with homothetic preferences, the WA can easily be violated when endowments are not proportional.⁴¹

In Section 2.F we offered a differential version of the WA for the case of demand functions. In a parallel fashion we can also do so for excess demand functions. It can be shown that a sufficient differential condition for the WA is

$$dp \cdot Dz(p) dp < 0 \text{ whenever } dp \cdot z(p) = 0 \text{ (i.e., whenever the price change is compensated) and } dp \text{ is not proportional to } p \text{ (i.e., relative prices change).} \quad (17.F.1)$$

Allowing for the first inequality to be weak, expression (17.F.1) constitutes also a necessary condition.⁴²

Under the homotheticity assumption, we have

$$D_{w_i} x_i(p, p \cdot \omega_i) = \frac{1}{p \cdot \omega_i} x_i(p, p \cdot \omega_i).$$

Denoting $S_i = S_i(p, p \cdot \omega_i)$, $x_i = x_i(p, p \cdot \omega_i)$, $\bar{x} = \sum_i x_i$ and $\bar{\omega} = \sum_i \omega_i$, this implies (recall 17.E.3)

$$\begin{aligned} Dz(p) &= \sum_i S_i(p, p \cdot \omega_i) - \sum_i \frac{1}{p \cdot \omega_i} x_i(p, p \cdot \omega_i) z_i(p, p \cdot \omega_i)^T \\ &= \sum_i S_i - \sum_i \frac{1}{p \cdot \omega_i} \left[x_i - \frac{p \cdot \omega_i}{p \cdot \bar{\omega}} \bar{x} \right] \left[x_i - \frac{p \cdot \omega_i}{p \cdot \bar{\omega}} \bar{x} \right]^T \\ &\quad + \sum_i \frac{1}{p \cdot \omega_i} \left[x_i - \frac{p \cdot \omega_i}{p \cdot \bar{\omega}} \bar{x} \right] \left[\omega_i - \frac{p \cdot \omega_i}{p \cdot \bar{\omega}} \bar{\omega} \right]^T - \frac{1}{p \cdot \bar{\omega}} \bar{x} z(p)^T. \end{aligned} \quad (17.F.2)$$

For any direction of price change dp with $dp \cdot z(p) = 0$, the first two terms on the right-hand side of equation (17.F.2) generate an effect of the appropriate sign [the $L \times L$ substitution matrices $S_i(p, p \cdot \omega_i)$ and variance matrices

$$-\left[x_i - \frac{p \cdot \omega_i}{p \cdot \bar{\omega}} \bar{x} \right] \left[x_i - \frac{p \cdot \omega_i}{p \cdot \bar{\omega}} \bar{x} \right]^T$$

are negative semidefinite], the fourth is null, *but the third is ambiguous*. It is quite possible for this covariance term to have the wrong sign (positive) and even for it to overcome the other two terms.⁴³ The situation to worry about is when $(1/(p \cdot \omega_i))x_i(p, p \cdot \omega_i)$ and ω_i are positively associated within the population of consumers; that is, when the consumers who consume (per dollar) more than the average (per dollar) consumption of some commodities tend to be those that are relatively well endowed (per dollar) with those commodities. It makes sense that this case will cause difficulties: If the price of a good increases, the consumers who are (net) sellers of the good (who are likely to be those relatively well endowed with it) experience a positive wealth effect, whereas the consumers who are (net) buyers experience a negative wealth effect. Hence, an increase in the total demand for the good will ensue if (net) sellers consume relatively more of the good (per dollar) than (net) buyers.

41. To reinforce this point, it is also worth mentioning that, in fact, if we are free to choose initial endowments, then the class of homothetic preferences imposes no restrictions on aggregate demand. Indeed, as we noted in Section 17.E, the basic conclusion of Proposition 17.E.2 can still be obtained with the further restriction that preferences be homothetic. See Mantel (1976) and the survey of Shafer and Sonnenschein (1982).

42. Suppose that $\Delta p \cdot z(p) = (p' - p) \cdot z(p) = 0$. Definition 17.F.1 implies then that $\Delta p \cdot \Delta z = (p' - p) \cdot (z(p') - z(p)) \leq 0$. Going to the differential limit and using the chain rule, it follows that $dp \cdot Dz(p) dp \leq 0$ whenever $dp \cdot z(p) = 0$.

43. But this cannot happen if the $x_i(p, p \cdot \omega_i)$ are collinear among themselves or if the ω_i are collinear among themselves. See Exercise 17.F.1.

Example 17.F.1: This is an example of a failure of the WA compatible with homotheticity and even with the property of gross substitution, which we will discuss shortly. Consider a four-commodity economy with two consumers. Consumer 1 has preferences and endowments for only the first two goods; that is, he has an excess demand function $z_1(p) = z_1(p_1, p_2)$ that does not depend on p_3 and p_4 and, further, is such that $z_{31}(p) = z_{41}(p) = 0$ for all p . Similarly, consumer 2 has preferences and endowments for only the last two goods.⁴⁴ We claim that if there is a price vector p' at which the excess demand of the two consumers is nonzero [i.e., $z_1(p') \neq 0$ and $z_2(p') \neq 0$], then the aggregate excess demand cannot satisfy the WA. To see this, choose $(\hat{p}_1, \hat{p}_2)$ and $(\hat{p}_3, \hat{p}_4)$ arbitrarily, except that $\hat{p}_1 z_{11}(p') + \hat{p}_2 z_{21}(p') < 0$ and $\hat{p}_3 z_{32}(p') + \hat{p}_4 z_{42}(p') < 0$. For $\alpha > 0$, take $q = (p'_1, p'_2, \alpha \hat{p}_3, \alpha \hat{p}_4)$ and $q' = (\alpha \hat{p}_1, \alpha \hat{p}_2, p'_3, p'_4)$. Then if $\alpha > 0$ is sufficiently large, we have $q \cdot z(q') < 0$ and $q' \cdot z(q) < 0$ (Exercise 17.F.2). ■

See Exercise 17.F.3 for yet another example.

Gross Substitution

We now investigate the implications of a condition of a different nature from the WA. We shall see that it yields a uniqueness result for situations that are reducible to formalization as exchange economies.

To motivate the concept (and justify its name), consider the demand function of a consumer in a two-commodity situation. At given prices the demand substitution matrix has negative diagonal entries and, as a consequence, positive off-diagonal entries: if the price of one good is raised, the compensated demand for the other good increases. However, if we do not net out the wealth effects (i.e., if we look at the effect of prices on uncompensated demand), then it is possible for an increase in the price of one good to decrease the demand of the two goods: in gross terms, the two goods may be complements. We say that the two goods are *gross substitutes* if this does not happen, that is, if an increase in the price of one good decreases the (uncompensated or gross) demand for that good and increases the (uncompensated or gross) demand for the other good. By extension, the same term is used in the L -commodity case for the property that asserts that when a price of one good increases, the demand of every other good increases (and, therefore, the demand of that good decreases). For $L > 2$, however, this is not by any means a necessary property of even compensated demand. In fact, the gross substitute property is very restrictive. Nonetheless, it can make sense for problems with a few very aggregated commodities or for those where commodities possess special symmetries (see Exercise 17.F.4).

Definition 17.F.2: The function $z(\cdot)$ has the *gross substitute* (GS) property if whenever p' and p are such that, for some ℓ , $p'_\ell > p_\ell$ and $p'_k = p_k$ for $k \neq \ell$, we have $z_k(p') > z_k(p)$ for $k \neq \ell$.

If, as is the case here, we are dealing with the aggregate excess demand of an economy, then the fact that $z(\cdot)$ is also homogeneous of degree zero has the consequence that with gross substitution we also have $z_\ell(p') < z_\ell(p)$ whenever p' and p are related as in Definition 17.F.2. To see this, let $\bar{p} = \alpha p$, where $\alpha = p'_\ell / p_\ell$. Note that $\bar{p}_\ell = p'_\ell$ and $\bar{p}_k > p'_k$ for $k \neq \ell$. Then the homogeneity of degree zero of $z(\cdot)$

44. Thus, this example can also be seen as a case of positive association between endowments and demands.

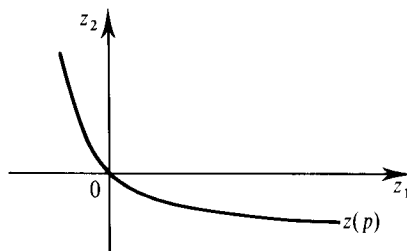


Figure 17.F.2

The offer curve of a gross substitute excess demand function.

tells us that $0 = z_\ell(\bar{p}) - z_\ell(p) = z_\ell(\bar{p}) - z_\ell(p') + z_\ell(p') - z_\ell(p)$. However, gross substitution implies that $z_\ell(\bar{p}) - z_\ell(p') > 0$ (change sequentially each price p'_k for $k \neq \ell$ to $\bar{p}_k$, applying the GS property at each step), and so $z_\ell(p') - z_\ell(p) < 0$.

The differential version of gross substitution is clear enough: At every p , it must be that $\partial z_k(p)/\partial p_\ell > 0$ for $k \neq \ell$; that is, the $L \times L$ matrix $Dz(p)$ has positive off-diagonal entries. In addition, when $z(\cdot)$ is an aggregate excess demand function, homogeneity of degree zero implies that $Dz(p)p = 0$, and so $\partial z_\ell(p)/\partial p_\ell < 0$ for all $\ell = 1, \dots, L$: the diagonal entries of $Dz(p)$ are all negative.

If in these definitions the inequalities are weak, one speaks of *weak gross substitution*.⁴⁵

Figure 17.F.2 represents the offer curve of a gross substitute excess demand function $L = 2$. As the relative price of good 1 increases, the excess demand for good 1 decreases and the excess demand for good 2 increases.

An important characteristic of the gross substitute property, which follows directly from its definition, is that it is *additive across excess demand functions*. In particular, if the individual excess demand functions satisfy it, then the aggregate function does also.

Example 17.F.2: Consider a utility function of the form $u_i(x_i) = \sum_\ell u_{\ell i}(x_{\ell i})$. If $-[x_{\ell i} u''_{\ell i}(x_{\ell i})/u'_{\ell i}(x_{\ell i})] < 1$ for all ℓ and $x_{\ell i}$, then the resulting excess demand function $z_i(p)$ has the gross substitute property for any initial endowments (Exercise 17.F.5). This condition is satisfied by $u_i(x_i) = (\sum_\ell \alpha_{\ell i} x_{\ell i}^\rho)^{1/\rho}$ for $0 < \rho < 1$ (Exercise 17.F.5). The limits of these preferences as $\rho \rightarrow 1$ and $\rho \rightarrow 0$ are preferences representable, respectively, by linear functions and by Cobb–Douglas utility functions (recall Exercise 3.C.6). As far as the gross substitution property is concerned, Cobb–Douglas preferences constitute a borderline case. Indeed, the excess demand function for good ℓ is then $z_{\ell i}(p) = \alpha_{\ell i}(p \cdot \omega_i)/p_\ell - \omega_{\ell i}$. If $\omega_{ki} > 0$, the excess demand for good

45. It is worth mentioning that functions satisfying the GS property arise naturally in many economic contexts. For example, if A is an $(L-1) \times (L-1)$ input–output matrix and $c \in \mathbb{R}_+^{L-1}$, then $c - (I - A)\alpha$ satisfies the (weak) GS property as a function of $\alpha \in \mathbb{R}_+^{L-1}$ (see Appendix A of Chapter 5 for the interpretation of these concepts). More generally, the equation system $g(\alpha) - \alpha$ associated with the fixed-point problem [i.e., find α such that $g(\alpha) = \alpha$] of an *increasing* function $g: \mathbb{R}_+^N \rightarrow \mathbb{R}_+^N$ [i.e., $g(\alpha) \geq g(\alpha')$ whenever $\alpha \geq \alpha'$] satisfies it (perhaps, again, in its weak version). Note that in these cases there is no homogeneity of degree zero or Walras' law—conditions specific to general equilibrium applications—to complement the GS property. This is significant because these conditions add substantially to the power of the GS property. See Exercise 17.F.16 for an exploration of the implications of the GS property without homogeneity of degree zero or Walras' law.

ℓ will respond positively to an increase in p_k . But if $\omega_{ki} = 0$, there will be no response.⁴⁶ ■

In the special case of exchange economies if the gross substitute property holds for aggregate excess demand then equilibrium is unique.

Proposition 17.F.3: An aggregate excess demand function $z(\cdot)$ that satisfies the gross substitute property has at most one exchange equilibrium; that is, $z(p) = 0$ has at most one (normalized) solution.

Proof: It suffices that we show that $z(p) = z(p')$ cannot occur whenever p and p' are two price vectors that are not collinear. By homogeneity of degree zero, we can assume that $p' \geq p$ and $p_\ell = p'_\ell$ for some ℓ . Now consider altering the price vector p' to obtain the price vector p in $L - 1$ steps, lowering (or keeping unaltered) the price of every commodity $k \neq \ell$ one at a time. By gross substitution, the excess demand of good ℓ cannot decrease in any step, and, because $p \neq p'$, it will actually increase in at least one step. Hence, $z_\ell(p) > z_\ell(p')$. ■

One might hope to establish uniqueness in economies with production by applying the GS property to the production inclusive excess demand $\bar{z}(\cdot)$. However, the direct use of the GS property in a production context is limited. Imagine, for example, a situation in which inputs and outputs are distinct goods. If the price of an input increases, the demand for every other input may decrease, not increase as the GS property would require, simply because the optimal level of output decreases. Indirectly, though, the gross substitute concept may still be quite helpful. Recall, in particular, that at the end of Section 17.B, we argued that it is always possible to reduce a production economy to an exchange economy in which, in effect, consumers exchange factor inputs and then engage in home production using a freely available constant returns technology. The aggregate excess demand in this derived exchange economy for factor inputs combines elements of both consumption and production and may well satisfy the GS property.⁴⁷

What is the relationship between gross substitution and the weak axiom? Clearly, the WA does not imply the GS property (the latter can be violated even in quasilinear, one-consumer economies). The converse relationship is not as obvious, but it is nevertheless true that the GS property does not imply the WA. In fact, Example 17.F.1, which violated the WA, could perfectly well satisfy GS.⁴⁸ There is, however, one connection that is important. The gross substitute property implies that

$$\text{If } z(p) = 0 \text{ and } z(p') \neq 0, \text{ then } p \cdot z(p') > 0. \quad (17.F.3)$$

We shall not prove condition (17.F.3) here. For the case in which $L = 2$, you are asked for a proof in Exercise 17.F.7. To understand (17.F.3), note that if p is the price vector of an

46. See also Grandmont (1992) for an interesting result where a Cobb–Douglas positive representative consumer, and therefore GS excess demand, is derived from a requirement that at any given price, the choice behavior is widely dispersed (in a certain precise sense) across consumers. Grandmont's is an example of a model in which the individual excess demand functions may not satisfy the gross substitute property but the aggregate function does.

47. See Mas-Colell (1991) and Exercise 17.F.6 for further elaborations on this point.

48. Therefore, in view of Proposition 17.F.1, we know that in a constant returns economy the fulfillment of the GS property by the excess demand of the consumers does *not* imply the uniqueness of equilibrium.

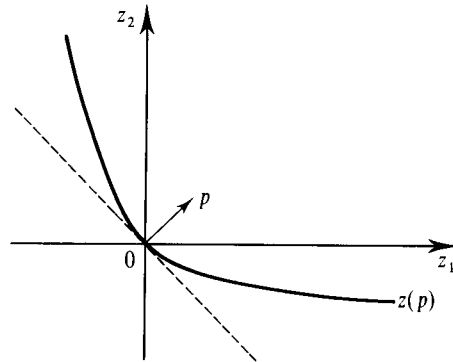


Figure 17.F.3
The revealed
preference property of
gross substitution.

(exchange) equilibrium and p' is not, then, since $z(p) = 0$, we have $p' \cdot z(p) = 0$, and therefore any nonequilibrium p' is revealed preferred to p . Hence, the requirement in (17.F.3) that $p \cdot z(p') > 0$ amounts to a restricted version of the WA asserting that no equilibrium price vector p can be revealed preferred to a nonequilibrium price vector p' . Geometrically, it says that the range of the excess demand function, $\{z(p') : p' \gg 0\} \subset \mathbb{R}^L$ (i.e., the offer curve), lies entirely above the hyperplane through the origin with normal vector p (see Figure 17.F.3). In parallel to Proposition 17.F.2, condition (17.F.3) implies the convexity of the equilibrium price set of the exchange economy, that is, of $\{p \in \mathbb{R}_{++}^L : z(p) = 0\} \subset \mathbb{R}^L$ (in Exercise 17.F.8, you are asked to show this). Interestingly, condition (17.F.3) is satisfied not only in the WA and the GS cases but also in the no-trade case, to be reviewed shortly.

In the differentiable case, there is a parallel way to explore the connection between the WA and gross substitution. Let $z(p) = 0$. The sufficient differential condition (17.F.1) for the WA tells us that $dp \cdot Dz(p) dp < 0$ for any dp not proportional to p . Suppose now that instead of the WA, we require that $Dz(p)$ has the gross substitute sign pattern. Because $z(p) = 0$, we have $p \cdot Dz(p) = 0$ and $Dz(p)p = 0$ [recall (17.E.1) and (17.E.2)]. Using these two properties it can then be shown that again we obtain $dp \cdot Dz(p) dp < 0$ for any dp not proportional to p (see Section M.D of the Mathematical Appendix). Hence, we can conclude that *at an exchange equilibrium price vector*, the GS property yields every local restriction implied by the WA. This is summarized in Proposition 12.F.4.

Proposition 17.F.4: If $z(\cdot)$ is an aggregate excess demand function, $z(p) = 0$, and $Dz(p)$ has the gross substitute sign pattern, then we also have $dp \cdot Dz(p) dp < 0$ whenever $dp \neq 0$ is not proportional to p .

Uniqueness as an Implication of Pareto Optimality

We now present a result that is not of great significance in itself but that is nonetheless interesting because it highlights a uniqueness implication of Pareto optimality. For simplicity, we restrict ourselves again to an exchange economy (see Exercise 17.F.9 for a generalization allowing for production).

Proposition 17.F.5: Suppose that the initial endowment allocation $(\omega_1, \dots, \omega_I)$ constitutes a Walrasian equilibrium allocation for an exchange economy with strictly convex and strongly monotone consumer preferences (i.e., no-trade is an equilibrium). Then this is the unique equilibrium allocation.

Proof: Let an allocation $x = (x_1, \dots, x_I)$ and price vector p constitute a Walrasian equilibrium when consumers' endowments are $(\omega_1, \dots, \omega_I)$. Since ω_i is affordable at

price vector p for each consumer i , we have $x_i \succeq_i \omega_i$ for all i . However, by the assumption of the proposition and the first welfare theorem, $(\omega_1, \dots, \omega_I)$ is a Pareto optimal allocation and so we must have $x_i \sim_i \omega_i$ for all i . But then we can conclude that $x_i = \omega_i$ for all i , because otherwise, by the strict convexity of preferences, the allocation $(\frac{1}{2}x_1 + \frac{1}{2}\omega_1, \dots, \frac{1}{2}x_I + \frac{1}{2}\omega_I)$ would be Pareto superior to $(\omega_1, \dots, \omega_I)$. ■

Index Analysis and Uniqueness (... and Nonuniqueness)

The index theorem (Proposition 17.D.2) provides a device to test for uniqueness in any given model. The idea is that if merely from the general maintained assumptions of the model we can attach a definite sign to the determinant of the Jacobian matrix of the equilibrium equations at any solution point, then the equilibrium must be unique. After all, the index theorem implies that sign uniformity across equilibria is impossible if there is multiplicity.

As a matter of fact, we could have proceeded by means of this index methodology for many of our previous uniqueness results. Take, for example, an exchange economy. In both the WA and the GS cases, whenever $z(p) = 0$, the matrix $Dz(p)$ is necessarily negative semidefinite [see the small-type discussion of expression (17.F.1) and Proposition 17.F.4]. Moreover, if an equilibrium is regular (i.e., if $\text{rank } Dz(p) = L - 1$), the negative semidefiniteness of $Dz(p)$ can be shown to imply that the index of the equilibrium is necessarily $+1$ (see Exercise 17.F.11). Hence, we can conclude that in both the WA and GS cases, any regular economy must have a unique (normalized) equilibrium price vector.

Although the index methodology provides a good research tool, it is often the case that, as here, uniqueness conditions lend themselves to direct proofs. It is a notable fact that some of the more subtle uses of index analysis are not to establish uniqueness but rather to establish nonuniqueness [the first usage of this type was made by Varian (1977)]. This is illustrated in Example 17.F.3.

Example 17.F.3: Suppose we have two one-consumer countries, $i = 1, 2$. Countries are symmetrically positioned relative to the home (H) and the foreign (F) good. To be specific, let each country have one unit of the home good as an endowment and none of the foreign good, and utility functions $u_i(x_{Hi}, x_{Fi}) = x_{Hi} - x_{Fi}^\rho$ for $-1 < \rho < 0$. Merely from symmetry considerations, it follows that there is a symmetric equilibrium $p = (1, 1)$. But we may be interested in knowing whether there are asymmetric equilibria. One way to proceed is as follows: compute the index of the symmetric equilibrium; a sufficient (but not necessary) condition for the existence of an asymmetric equilibrium is that this index be negative (i.e., -1).⁴⁹ If we carry out the computation for the present example (you are asked to do so in Exercise 17.F.13), we see that the index is negative if at prices $p = (1, 1)$ the wealth effects in each country are so biased toward the home good that an increase in the price of the good of country 1, say, actually increases the demand for this good in country 1 by more than it decreases the demand from country 2. ■

49. In this, as typically in any example, the excess demand function fails to be differentiable at prices at which demand just “hits” the boundary. Typically (we *could* say “generically”), these prices will not be equilibrium prices and the validity of the index theorem is not affected by these nondifferentiabilities.

17.G Comparative Statics Analysis

Comparative statics is the analytical methodology that concerns itself with the study of how the equilibria of a system are affected by changes (often described as “shocks”) in various environmental parameters. In this section, we examine the comparative static properties of Walrasian equilibria.

To be concrete, we consider an exchange economy formalized by a system of aggregate excess demand equations for the first $L - 1$ commodities:

$$\hat{z}(p; q) = (z_1(p; q), \dots, z_{L-1}(p; q)).$$

Here, $q \in \mathbb{R}^N$ is a vector of N parameters influencing preferences or endowments (or both). Throughout, we normalize $p_L = 1$.

Suppose the value of the parameters is given initially by the vector $\bar{q}$ and that $\bar{p}$ is an equilibrium price vector for $\bar{q}$; that is, $\hat{z}(\bar{p}; \bar{q}) = 0$. We wish to analyze the effect of a shock in the exogenous parameters q on the endogenous variable p solving the system. A first difficulty for doing so is the possibility of multiplicity of equilibrium: the system of $L - 1$ equations in $L - 1$ unknowns $\hat{z}(\cdot; q) = 0$ may have more than one solution for the relevant values of q , and thus we may need to decide which equilibrium to single out after a shock.

If the change in the values of the parameters from $\bar{q}$ is small, then a familiar approach to this problem is available. It consists of focusing on the *local* effects on p , that is, on the solutions that remain near $\bar{p}$. Assuming the differentiability of $\hat{z}(p; q)$, we may determine those effects by applying the implicit function theorem (see Section M.E of the Mathematical Appendix). Indeed, if the system $\hat{z}(\cdot; \bar{q}) = 0$ is regular at the solution $\bar{p}$, that is, if the $(L - 1) \times (L - 1)$ matrix $D_p \hat{z}(\bar{p}; \bar{q})$ has rank $L - 1$,⁵⁰ then for a neighborhood of $(\bar{p}; \bar{q})$ we can express the equilibrium price vector as a function $p(q) = (p_1(q), \dots, p_{L-1}(q))$ whose $(L - 1) \times N$ derivative matrix at $\bar{q}$ is

$$Dp(\bar{q}) = -[D_p \hat{z}(\bar{p}; \bar{q})]^{-1} D_q \hat{z}(\bar{p}; \bar{q}). \quad (17.G.1)$$

What can we say about the first-order effects $Dp(\bar{q})$? Expression (17.G.1) and Proposition 17.E.2 [which told us that the matrix of price effects $D_p \hat{z}(\bar{p}; \bar{q})$ is unrestricted when $I \geq L$] strongly suggest that, without further assumptions, the “anything goes” principle applies to the comparative statics of equilibrium in the same manner that in Section 17.E it applied to the closely related issue of the effects of price changes on excess demand. We now elaborate on this point in the context of a specific example.

Let the list of parameters under consideration be the vector $\hat{\omega}_1 = (\omega_{11}, \dots, \omega_{L-1,1})$ of initial endowments of the first consumer for the first $L - 1$ commodities. All of the remaining endowments are kept fixed. As before we assume that $\hat{z}(\cdot; \hat{\omega}_1) = 0$ is regular at the solution $\bar{p}$. It can be shown (see Exercise 17.G.1) that if the demand function of the first consumer satisfies a strict normality condition, then $\text{rank } Dp(\hat{\omega}_1) = L - 1$, where $p(\cdot)$ is the locally defined solution function with $p(\hat{\omega}_1) = \bar{p}$. Proposition 17.G.1 tells us that if there are enough consumers then this is all that we can say.

50. In a slight abuse of notation, we let $D_p \hat{z}(\bar{p}; \bar{q})$ stand for the matrix obtained from $D_p \hat{z}(\bar{p}; \bar{q})$ by deleting the last row and column.

Proposition 17.G.1: Given any price vector $\bar{p}$, endowments for the first consumer of the first $L - 1$ commodities $\hat{\omega}_1 = (\bar{\omega}_{11}, \dots, \bar{\omega}_{L-1,1})$, and a $(L - 1) \times (L - 1)$ nonsingular matrix B , there is an exchange economy formed by $L + 1$ consumers in which the first consumer has the prescribed endowments of the first $L - 1$ commodities, $\hat{z}(\bar{p}; \hat{\omega}_1) = 0$, $\hat{z}(\cdot, \hat{\omega}_1) = 0$ is regular at $\bar{p}$ and $Dp(\hat{\omega}_1) = B$.

Proof: Let the first consumer have endowments with the prescribed amounts of the first $L - 1$ commodities, and give to this consumer arbitrary preferences, with the single restriction that $D_{\omega_1} \hat{z}_1(\bar{p}; \hat{\omega}_1)$ be nonsingular (it suffices for this that the demand function of consumer 1 satisfies a strict normality condition; again see Exercise 17.G.1). Since $D_{\omega_1} \hat{z}_1(\bar{p}; \hat{\omega}_1) = D_{\omega_1} \hat{z}_1(\bar{p}; \hat{\omega}_1)$, expression (17.G.1) tells us that we are looking for an additional collection of L consumers such that the resulting $(L + 1)$ -consumer economy has $\hat{z}(\bar{p}; \hat{\omega}_1) = 0$ and

$$D_p \hat{z}(\bar{p}; \hat{\omega}_1) = -D_{\omega_1} \hat{z}_1(\bar{p}; \hat{\omega}_1) B^{-1} \quad (17.G.2)$$

Note that the $(L - 1) \times (L - 1)$ matrix defined in (17.G.2) is nonsingular. Thus, we have reduced our problem to the following: can we find L consumers whose aggregate excess demand at $\bar{p}$ is $-\hat{z}_1(\bar{p}; \hat{\omega}_1)$ and whose aggregate $(L - 1) \times (L - 1)$ matrix of price effects is $\hat{A} = -D_{\omega_1} \hat{z}_1(\bar{p}; \hat{\omega}_1) B^{-1} - D_p \hat{z}_1(\bar{p}; \hat{\omega}_1)$? It follows from Proposition 17.E.2 that the answer to this question is “yes” (note that the restrictions that Proposition 17.E.2 imposes on the $L \times L$ matrix A place no restriction on the matrix obtained by deleting one row and one column of A). ■

Proposition 17.G.1 shows that any first-order effect is possible. As in Section 17.E (recall Figure 17.E.3), it is also the case here that if there are prior restrictions on initial endowments and if consumption must be nonnegative, then there are again comparative statics restrictions of a global character. [See Brown and Matzkin (1993) for a recent investigation of this point.]

There are a number of comparative static effects that, ideally, we would like to have and that seem economically intuitive: For example, that if the endowment of one good increases, then its equilibrium price decreases. Nevertheless, strong conditions are required for them to hold. By now this should not surprise us: We already know that wealth effects and/or the lack of sufficient substitutability can undermine intuitive comparative static effects. The latest instance we have seen of this occurring has been precisely Proposition 17.G.1.

The analysis of uniqueness in Section 17.E may lead us to suspect that good comparative statics effects can hold if aggregate excess demand satisfies either weak-axiom-like conditions (recall Definition 17.F.1) or gross substitution properties (see Definition 17.F.2). This is in fact so. We consider first the implications of a weak-axiom-like restriction on aggregate excess demand.

Proposition 17.G.2: Suppose that $\hat{z}(\bar{p}; \bar{q}) = 0$, where $\hat{z}(\cdot)$ is differentiable. If $D_q \hat{z}(\bar{p}; \bar{q})$ is negative definite,⁵¹ then

$$(D_q \hat{z}(\bar{p}; \bar{q}) dq) \cdot (Dp(\bar{q}) dq) \geq 0 \text{ for any } dq. \quad (17.G.3)$$

51. This condition is independent of which particular commodity has been labeled as L (see Section M.D of the Mathematical Appendix).

Proof: The inverse of a negative definite matrix is negative definite. Therefore $[D_q \hat{z}(\bar{p}; \bar{q})]^{-1}$ is negative definite (see Section M.D of the Mathematical Appendix). Hence, by (17.G.1) we have

$$(D_q \hat{z}(\bar{p}; \bar{q}) dq) \cdot (D_p(\bar{q}) dq) = -D_q \hat{z}(\bar{p}; \bar{q}) dq \cdot [D_p \hat{z}(\bar{p}; \bar{q})]^{-1} D_q \hat{z}(\bar{p}; \bar{q}) dq \geq 0$$

which is precisely (17.G.3). ■

The weak axiom implies the negative semidefiniteness of $D_p \hat{z}(\bar{p}; \bar{q})$ whenever $\hat{z}(\bar{p}; \bar{q}) = 0$ [see expression (17.F.1) and the remark following it]. Therefore, the assumption of Proposition 17.G.2 amounts to a small strengthening of this implication. Its conclusion says that for any infinitesimal shock dq in q , the induced shock to excess demand at prices fixed at $\bar{p}$, $D_q \hat{z}(\bar{p}; \bar{q}) dq$, and the induced shock in equilibrium prices, $D_p \hat{z}(\bar{p}; \bar{q}) dq$, move “in the same direction” (more precisely, as vectors in $\mathbb{R}^{L-1}$ they form an acute angle). For example, a shock that at fixed prices affects only the aggregate excess demand of the first good,⁵² say by decreasing it, will necessarily decrease the equilibrium price of this good. Note that this does *not* say that if ω_{11} increases then the equilibrium price of good 1 decreases. Under an assumption of normal demand, this change in ω_{11} does indeed decrease the excess demand for good 1 at $\bar{p}$ but it also affects the excess demand for all other goods (see Exercise 17.G.2).

We next consider in Proposition 17.G.3 the implications of gross substitution (or, more precisely, of gross substitution holding locally at $(\bar{p}; \bar{q})$).

Proposition 17.G.3: Suppose that $\hat{z}(\bar{p}; \bar{q}) = 0$, where $\hat{z}(\cdot; \cdot)$ is differentiable. If the $L \times L$ matrix $D_p \hat{z}(\bar{p}; \bar{q})$ has negative diagonal entries and positive off-diagonal entries, then $[D_p \hat{z}(\bar{p}; \bar{q})]^{-1}$ has all its entries negative.

Proof: Because of the homogeneity of degree zero of excess demand (recall Exercise 17.E.1), we have $D_p \hat{z}(\bar{p}; \bar{q}) \bar{p} = 0$, and so $D_p \hat{z}(\bar{p}; \bar{q}) \hat{p} \ll 0$, where $\hat{p} = (\bar{p}_1, \dots, \bar{p}_{L-1})$. Denote by I the $(L-1) \times (L-1)$ identity matrix and take an $r > 0$ large enough for the matrix $A = (1/r) D_p \hat{z}(\bar{p}; \bar{q}) + I$ to have all its entries positive. Then $D_p \hat{z}(\bar{p}; \bar{q}) = -r[I - A]$, and therefore $D_p \hat{z}(\bar{p}; \bar{q}) \hat{p} \ll 0$ yields $(I - A) \hat{p} \gg 0$; that is, the positive matrix A , viewed formally as an input–output matrix, is productive (see Appendix A of Chapter 5; the fact that the diagonal entries of A are not zero is inessential). Hence, as we showed in the proof of Proposition 5.AA.1, the matrix $[I - A]^{-1}$ exists and has all its entries positive. From $[D_p \hat{z}(\bar{p}; \bar{q})]^{-1} = -(1/r)[I - A]^{-1}$ we have our conclusion. ■

It follows from Proposition 17.G.3 and expression (17.G.1) that, given gross substitution, if $D_q \hat{z}(\bar{p}; \bar{q}) dq \ll 0$, that is, if the excess demand for all of the first $L-1$ goods decreases as a consequence of the shock (and therefore the excess demand for the L th good increases), then $D_p \hat{z}(\bar{p}; \bar{q}) dq \ll 0$. That is, the equilibrium prices of the

52. What this means is that the excess demand of good 2 to $L-1$ is not changed. By Walras' law, the excess demand of good L must change.

first $L - 1$ goods (relative to the price of the L th good) decrease.⁵³ In particular, suppose again that consumer 1's initial endowment of some good decreases. By labelling commodities appropriately, we can let this good be commodity L . Under the assumption of normal demand for consumer 1, a decrease in ω_{L1} , at the fixed price vector $\bar{p}$, will decrease the excess demand for the first $L - 1$ goods. Therefore, the prices of the first $L - 1$ goods decrease and so we now reach the conclusion that we could not obtain by means of Proposition 17.G.2: if the endowments of a single good decrease then its price (relative to the price of any other good) increases. This suggests, incidentally, that the assumptions of Proposition 17.G.3 are strictly stronger than those of Proposition 17.G.2. Indeed, as we saw in Proposition 17.F.4, if $z(\bar{p}; \bar{q}) = 0$ and the $L \times L$ matrix $D_p z(\bar{p}; \bar{q})$ satisfies the gross substitute property, then $dp \cdot D_p z(\bar{p}; \bar{q}) dp < 0$ whenever $dp \neq 0$ is not proportional to $\bar{p}$. In particular, by letting $dp_L = 0$ we have that the matrix $D_p \hat{z}(\bar{p}; \bar{q})$ is negative definite.

Expression (17.G.1) allows us to explicitly compute the effects of an infinitesimal shock. In fact, it also offers a practical computational method to estimate the local effects of small (but perhaps not infinitesimal) shocks. Suppose that the value of the vector of parameters after the shock is $\bar{q}$ and, for $t \in [0, 1]$, consider a continuous function $\hat{z}(\cdot, t)$ that, as t ranges from $t = 0$ to $t = 1$, distorts $\hat{z}(\cdot; \bar{q})$ into $\hat{z}(\cdot; \bar{q})$. An example of such a function, called a *homotopy*, is

$$\hat{z}(\cdot, t) = (1 - t)\hat{z}(\cdot; \bar{q}) + t\hat{z}(\cdot; \bar{q}).$$

Denote the solution set by $E = \{(t, p) : \hat{z}(p, t) = 0\}$. Then we may attempt to determine $p(\bar{q})$ by following a segment in the solution set that starts at $(0, \bar{p})$.⁵⁴ If $\bar{q}$ is close to $\bar{q}$, and the initial situation $\bar{p}$ is regular, then we are in the simple case of Figure 17.G.1(a): there is a unique segment that connects $(0, \bar{p})$ to some $(1, \bar{p})$.⁵⁵ Naturally, we then put $p(\bar{q}) = \bar{p}$.

If $\bar{q}$ is not close to $\bar{q}$ but nevertheless $\hat{z}(\cdot, t)$ is a regular excess demand function for every t [this will be the case if, for example, $z(\cdot, t)$ satisfies, for every t , any of the uniqueness conditions covered in Section 17.F], then this procedure will still succeed in going from $t = 0$ to $t = 1$ and, therefore, in determining an equilibrium for $\bar{q}$.⁵⁶ Unfortunately, if the shock is large, we can easily find ourselves in situations such as Figures 17.G.1(b) and 17.G.1(c), where at some t' the economy $\hat{z}(\cdot, t')$ is not regular and at $(t', p_{t'})$ there is no natural

53. This conclusion holds for nonlocal shocks as well. To see this let $Dz(p; q)$ have the gross substitute sign pattern throughout its domain and suppose that $\hat{z}(p; \bar{q}) \ll \hat{z}(p; \bar{q})$ for all p . For $t \in [0, 1]$, define $\hat{z}(p; t) = t\hat{z}(p; \bar{q}) + (1 - t)\hat{z}(p; \bar{q})$. Denote by $p(t)$ the solution to $\hat{z}(p; t) = 0$. Note that $D_t \hat{z}(p(t); t) dt = \hat{z}(p(t); \bar{q}) - \hat{z}(p(t); \bar{q}) \ll 0$ for all t and therefore, by Proposition 17.G.3, $Dp(t) dt \ll 0$ for all t . But then, for any $\ell = 1, \dots, L - 1$, we have

$$p_\ell(\bar{q}) - p_\ell(\bar{q}) = \int_0^1 \left[\frac{\partial p_\ell(t)}{\partial t} \right] dt < 0.$$

In Exercise 17.G.3 you can find a more direct approach to the global theory. See also Milgrom and Shannon (1994) for much more on the latter approach.

54. In practice, "following" a segment involves the application of appropriate numerical techniques; see Garcia-Zangwill (1981), Kehoe (1991), and references therein.

55. Moreover, if the shock is sufficiently small, the $\bar{p}$ so obtained is independent of the particular homotopy used.

56. However, if there are multiple equilibria at $\bar{q}$, then which equilibrium we find may now depend on the homotopy.

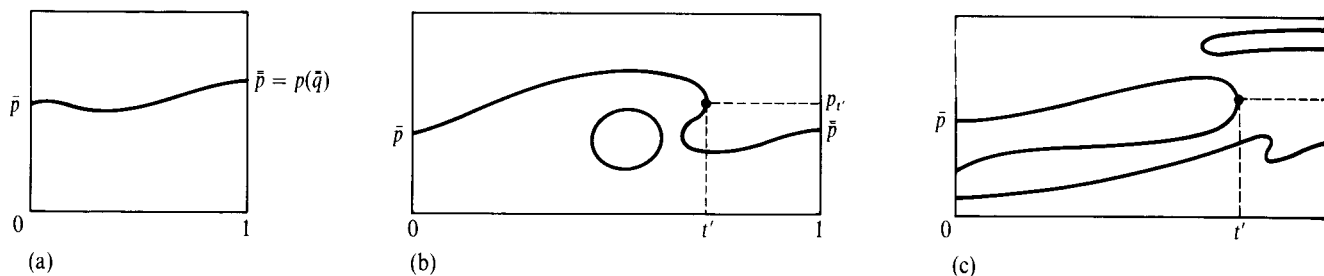


Figure 17.G.1 Comparative statics in the large: the general case.

continuation of the path as t increases.⁵⁷ To obtain an equilibrium $\bar{p}$ for $\bar{q}$ there is then no real alternative but to appeal to general algorithms for the solution of the system of equations $\hat{z}(\cdot; \bar{q}) = 0$. It is a sobering thought that which solution we come up with at $\bar{q}$ may be dictated more by our numerical technology than by our initial position $(\bar{p}; \bar{q})$. This is most unsatisfactory, and it is a manifestation of a serious shortcoming—the lack of a theory of equilibrium selection.

17.H Tâtonnement Stability

We have, so far, carried out an extensive analysis of equilibrium equations. A characteristic feature that distinguishes economics from other scientific fields is that, for us, the equations of equilibrium constitute the center of our discipline. Other sciences, such as physics or even ecology, put comparatively more emphasis on the determination of dynamic laws of change. In contrast, up to now, we have hardly mentioned dynamics. The reason, informally speaking, is that economists are good (or so we hope) at recognizing a state of equilibrium but are poor at predicting precisely how an economy in disequilibrium will evolve. Certainly there are intuitive dynamic principles: if demand is larger than supply then the price will increase, if price is larger than marginal cost then production will expand, if industry profits are positive and there are no barriers to entry, then new firms will enter, and so on. The difficulty is in translating these informal principles into precise dynamic laws.⁵⁸

The most famous attempt at this translation was made by Walras (1874), and the modern version of his ideas have come to be known as the theory of *tâtonnement stability*. In this section, we review two *tâtonnement*-style models, one of pure price adjustment and the other of pure quantity adjustment. We should emphasize,

57. Note that by reversing the direction of change of t we can continue to move along the segments in these two figures (this is actually quite a general fact). If $\bar{p}$ is the only solution at $t = 0$, as in 17.G.1(b), then the segment necessarily ends with a $(1, \bar{p})$. Thus, in some sense we have succeeded in finding an equilibrium for $\bar{q}$ that is associated with our initial $\bar{p}$. But the association is very weak: it may depend on the particular homotopy and it requires the parameter-reversal procedure. If, as in Figure 17.G.1(c), $\bar{p}$ is not the only equilibrium at $t = 0$, then the procedure may simply not work: the segment that starts at $(0, \bar{p})$ goes back to $t = 0$.

58. Refer to Hahn (1982) for a general review.

however, that those are just two examples. Indeed, one of the difficulties in this area is the plethora of plausible disequilibrium models. Although there is a single way to be in equilibrium, there are many different ways to be in disequilibrium.

Price Tâtonnement

We consider an exchange economy formalized by means of an excess demand function $z(\cdot)$. Suppose that we have an initial p that is not an equilibrium price vector, so that $z(p) \neq 0$. For example, the economy may have undergone a shock and p may be the preshock equilibrium price vector. Then the demand-and-supply principle suggests that prices will adjust upward for goods in excess demand and downward for those in excess supply. This is what was proposed by Walras; in a differential equation version put forward by Samuelson (1947), it takes the specific form

$$\frac{dp_\ell}{dt} = c_\ell z_\ell(p) \quad \text{for every } \ell, \quad (17.H.1)$$

where dp_ℓ/dt is the rate of change of the price for the ℓ th good and $c_\ell > 0$ is a constant affecting the speed of adjustment.

Simple as (17.H.1) is, its interpretation is fraught with difficulties. Which economic agent is in charge of prices? For that matter, why must the “law of one price” hold out of equilibrium (i.e., why must identical goods have identical prices out of equilibrium)? What sort of time does “ t ” represent? It cannot possibly be *real* time because, as the model stands, a disequilibrium p is not compatible with feasibility (i.e., not all consumption plans can be simultaneously realized).

Perhaps the most sensible answer to all these questions is that (17.H.1) is best thought of not as modeling the actual evolution of a demand-and-supply driven economy, but rather as a tentative trial-and-error process taking place in fictional time and run by an abstract market agent bent on finding the equilibrium level of prices (or, more modestly, bent on restoring equilibrium after a disturbance).⁵⁹

The hope is that, in spite of its idealized nature, the analysis of (17.H.1) will provide further insights into the properties of equilibria. Even perhaps some help in distinguishing good from poorly behaved equilibria. The analysis is at its most suggestive in the two-commodity case. For this case, Figure 17.H.1 represents the excess demand of the first good as a function of the relative price p_1/p_2 . The actual dynamic trajectory of relative prices depends both on the initial levels of absolute prices and on the differential price changes prescribed by (17.H.1).⁶⁰ But note that, whatever the initial levels of absolute prices, $p_1(t)/p_2(t)$ increases at t if and only if $z_1(p_1(t)/p_2(t), 1) > 0$. In Figure 17.H.1 we see the following two features of the adjustment equations (17.H.1).

(a) Call an equilibrium $(\bar{p}_1, \bar{p}_2)$ *locally stable* if, whenever the initial price vector is sufficiently close to it, the dynamic trajectory causes relative prices to converge to the equilibrium relative prices $\bar{p}_1/\bar{p}_2$ (the equilibrium is *locally totally unstable* if any

59. This is, in essence, the idea of Walras (tâtonnement means “groping” in French), who took inspiration from the functioning of the auctioneer-directed markets of the Paris stock exchange. The idea was made completely explicit by Barone (1908) and by Lange (1938), who went so far as to propose the tâtonnement procedure as an actual computing device for a centrally planned economy.

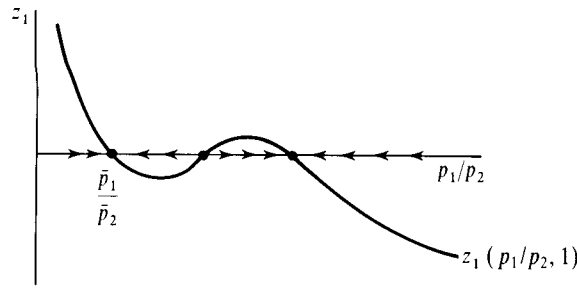


Figure 17.H.1
Tâtonnement
trajectories for $L = 2$.

disturbance leads the relative prices to diverge from $\bar{p}_1/\bar{p}_2$). Then a (regular) equilibrium $\bar{p}_1/\bar{p}_2$ is *locally stable* or *locally totally unstable* according to the sign of the slope of excess demand at the equilibrium, that is, according to the index of the equilibrium (recall Definition 17.D.2). If excess demand slopes downward at $\bar{p}_1/\bar{p}_2$ (as in Figure 17.H.1), then a slight displacement of p_1/p_2 above $\bar{p}_1/\bar{p}_2$ will generate excess supply for good 1 (and excess demand for good 2), and therefore the relative price will move back toward the equilibrium level $\bar{p}_1/\bar{p}_2$. The effect is the reverse if excess demand slopes upward at $\bar{p}_1/\bar{p}_2$.

(b) There is *system stability*, that is, for any initial position $(p_1(0), p_2(0))$, the corresponding trajectory of relative prices $p_1(t)/p_2(t)$ converges to some equilibrium arbitrarily closely as $t \rightarrow \infty$.

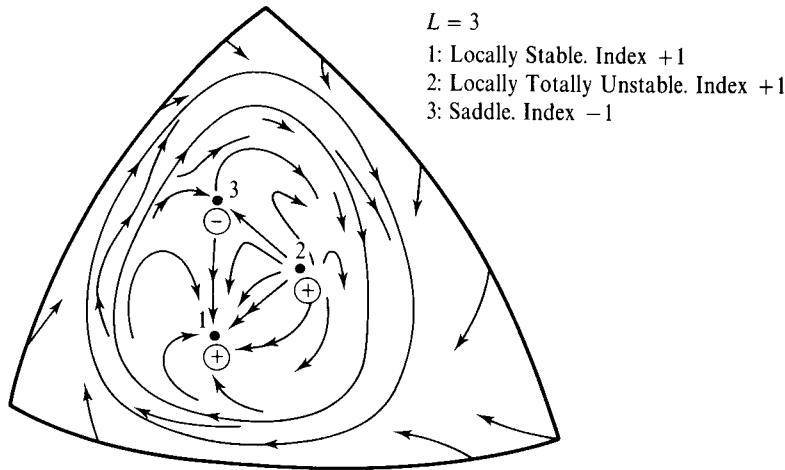
For regular, two-commodity, economies, properties (a) and (b) give a complete picture of the dynamics. It is very satisfactory picture that accounts for the persistency of tâtonnement stability analysis: a theory yielding properties (a) and (b) must be saying something with economic content.

Unfortunately, as soon as $L > 2$ neither the local conclusions (a) nor the global conclusions (b) of the two-commodity case generalize. This should not surprise us, since the price dynamics in (17.H.1) are entirely driven by the excess demand function, and we know (Propositions 17.E.2 and 17.E.3) that the latter is not restricted in any way (beyond the boundary conditions). Consider an example for $L = 3$ and $c_1 = c_2 = c_3 = 1$. In Figure 17.H.2 we represent the normalized set of prices $S = \{p \gg 0 : (p_1)^2 + (p_2)^2 + (p_3)^2 = 1\}$. This normalization has the virtue that, for any excess demand function $z(p)$, the dynamic flow $p(t)$ generated by the differential equation $dp_\ell/dt = z_\ell(p)$, $\ell = 1, 2, 3$, remains in S [i.e., if $p(0) \in S$ then $p(t) \in S$ for all t]. This is a consequence of Walras' law:

$$\frac{d(p_1(t)^2 + p_2(t)^2 + p_3(t)^2)}{dt} = 2p_1(t)z_1(p(t)) + 2p_2(t)z_2(p(t)) + 2p_3(t)z_3(p(t)) = 0.$$

Thus, the dynamics of p can be represented by trajectories in S , the direction vector of the trajectory at any $p(t)$ being the direction of the excess demand vector $z(p(t))$. We conclude, therefore, that the only restrictions on the trajectories imposed by the general theory are those derived from the boundary behavior of excess demand. In Figure 17.H.2 we represent a possible field of trajectories. In the figure, when the

60. Note that although the change in p_ℓ at t prescribed by (17.H.1) depends only on the relative prices p_1/p_2 for $\ell = 1, 2$, the change in the price ratio p_1/p_2 at t depends both on the current price ratio and on the current absolute levels of p_1 and p_2 .

**Figure 17.H.2**

An example of tâtonnement trajectories for $L = 3$.

price of a good goes to zero the excess demand for the good becomes positive (thus, in particular, the trajectories point inward near the boundary). However, properties (a) and (b) are both violated: There are (regular) equilibria that are neither locally stable nor locally totally unstable (they are “saddle points,” such as the equilibrium labeled 3 in the figure), and from some initial positions prices may not converge to any equilibrium.⁶¹

In a more positive spirit, we now argue that for the cases where we have succeeded in proving the uniqueness of Walrasian equilibrium, we are also able to establish the convergence of any price trajectory to this equilibrium (this property is called *global stability*).⁶² The next proposition covers, in particular, the weak axiom, the gross substitute, and the no-trade cases studied in Section 17.F.⁶³ These three cases have in common that they satisfy the weak axiom when we restrict ourselves to comparisons between equilibrium and nonequilibrium prices [see the discussion of condition (17.F.3) in Section 17.F]. That is, for the unique (normalized) equilibrium price vector p^* arising in these cases we have: “If $z(p^*) = 0$ then $p^* \cdot z(p) > 0$ for any p not proportional to p^* .”

Proposition 17.H.1: Suppose that $z(p^*) = 0$ and $p^* \cdot z(p) > 0$ for every p not proportional to p^* . Then the relative prices of any solution trajectory of the differential equation (17.H.1) converge to the relative prices of p^* .

Proof: Consider the (Euclidean) distance function $f(p) = \sum_{\ell} (1/c_{\ell})(p_{\ell} - p_{\ell}^*)^2$. For any trajectory $p(t)$ let us then focus on the distance $f(p(t))$ at points t along the trajectory. We have

61. We should warn against deriving any comfort when prices converge to a limit cycle. Recall that this price tâtonnement is not happening in real time. The dynamic analysis has a hope of telling us something significant only if it converges.

62. Warning: uniqueness by itself does not imply stability—except for $L = 2$. You should try to draw a counterexample in the style of Figure 17.H.2.

63. For a proof specific to the gross substitute case, see Exercise 17.H.1.

$$\begin{aligned}
\frac{df(p(t))}{dt} &= 2 \sum_{\ell} \frac{1}{c_{\ell}} (p_{\ell}(t) - p_{\ell}^*) \frac{dp_{\ell}(t)}{dt} \\
&= \sum_{\ell} \frac{1}{c_{\ell}} (p_{\ell}(t) - p_{\ell}^*) c_{\ell} z_{\ell}(p(t)) \\
&= -p^* \cdot z(p(t)) \leq 0,
\end{aligned}$$

where the last inequality is strict if and only if $p(t)$ is not proportional to p^* . We conclude that the price vector $p(t)$ monotonically approaches the price vector p^* [in fact, since the same argument applies to αp^* , $p(t)$ must be monotonically approaching any αp^*]. This does not mean that $p(t)$ reaches a vicinity of p^* . Typically it will not: the rate of approach of $p(t)$ to p^* will go to zero before $p(t)$ gets near p^* . But the rate of approach can go to zero only if $p(t)$ becomes nearly proportional to p^* as $t \rightarrow \infty$, in which case the relative prices do converge.⁶⁴ ■

We can gain further insight into the dynamics of tâtonnement by carrying out a local analysis. It will be more convenient now if we fix $p_L = 1$ and, consequently, we limit (17.H.1) to the first $L - 1$ coordinates. Accordingly we denote $\hat{z}(p) = (\hat{z}_1(p), \dots, \hat{z}_{L-1}(p))$. Suppose that $\hat{z}(p^*) = 0$. A standard result of differential equation theory tells us that if the $(L - 1) \times (L - 1)$ matrix $D\hat{z}(p^*)$ is nonsingular (i.e., if the equilibrium is regular), then the behavior of the trajectories in a neighborhood of p^* is controlled by the linearization of the system at p^* , that is, by $CD\hat{z}(p^*)$, where C is the $(L - 1) \times (L - 1)$ diagonal matrix whose ℓ th diagonal entry is the constant c_{ℓ} . One says that p^* is *locally stable* if there is $\varepsilon > 0$ such that $p(t) \rightarrow p^*$ whenever $\|p(0) - p^*\| < \varepsilon$ (i.e., for small perturbations the equilibrium will tend to restore itself). It then turns out that p^* is locally stable if and only if all the eigenvalues of $CD\hat{z}(p^*)$ have negative real parts. In addition, p^* is locally stable irrespective of the speeds of adjustment⁶⁵ (i.e., for all positive diagonal matrices C) if $D\hat{z}(p^*)$ is negative definite (see Section M.D of the Mathematical Appendix).⁶⁶

One way to understand why the previous local stability result for the tâtonnement dynamics requires strong conditions on $D\hat{z}(p^*)$ is to note that we are in fact imposing the condition that the price of a commodity reacts *only* to the excess demand or supply for the same commodity. An ideal market agent may want to adjust these prices with an eye also to the effects of the adjustment on the excess demand for the *other* commodities. One concrete possibility is the following: if excess demand at time t is $\hat{z}(p(t))$, then the market agent adjusts prices by some amount $dp/dt = (dp_1/dt, \dots, dp_{L-1}/dt)$ so as to cause a *proportional* decrease in the magnitude of all excess demands and supplies. That is, $D\hat{z}(p(t))(dp/dt) = -\lambda\hat{z}(p(t))$ for some $\lambda > 0$, or, if the relevant inverse exists,

$$\frac{dp}{dt} = -\lambda[D\hat{z}(p)]^{-1}\hat{z}(p) \quad (17.H.2)$$

This adjustment equation is known as *Newton's method* and is a standard technique of numerical analysis. If $D\hat{z}(p^*)$ is nonsingular, so that $[D\hat{z}(p^*)]^{-1}$ exists, then (17.H.2) always

64. Continuous real-valued functions that take decreasing values along any dynamic trajectory and the value zero only at stationary points are known as *Lyapunov functions*.

65. How could we pretend to know much about speeds of adjustments?

66. Note that this fits nicely with Proposition 17.H.1 because the revealed-preference-like property postulated there implies the negative (semi)definiteness of $D\hat{z}(p)$ at the equilibrium price vector p^* .

succeeds in restoring equilibrium after a small disturbance. Thus we see the contrast: for tâtonnement stability, we impose few informational restrictions on the adjustment process [to determine the change in p we only need to know $\hat{z}(p)$; in particular, no knowledge of the derivatives of $\hat{z}(\cdot)$ is required], but convergence is guaranteed only in special circumstances. For the Newton method, local convergence always obtains, but to determine the directions of price change at any p we need to know all the excess demands $\hat{z}(p)$ and all the price effects $D\hat{z}(p)$. See Smale (1976) and Saari and Simon (1978) for classic contributions to this type of Newton price dynamics.

Quantity Tâtonnement

In the analysis so far, prices could be out of equilibrium but quantities, that is to say the amounts demanded and supplied, are always at their equilibrium (i.e., utility and profit-maximizing) values. We now briefly consider a model in which quantities rather than prices may be in disequilibrium.⁶⁷ This is best done in a production context.

To be very concrete, suppose that there is a single production set Y .⁶⁸ At any moment of time, we assume that there is given a single, fixed production vector $y \in Y$. Prices, however, are always in equilibrium in the sense that the general equilibrium system of the economy, conditional on y , generates some equilibrium price system $p(y)$ (that is to say, we proceed in the short run as if the short-run production set were $\{y\} - \mathbb{R}_+^L$). This describes the short-run equilibrium of the economy.

What is an appropriate dynamics for this economy? It makes sense to think that, whatever it is, the change in production at time t , $dy(t)/dt \in \mathbb{R}^L$, moves production in a direction that increases profits *when the price vector at time t , $p(y(t))$, is taken as given*:

Definition 17.H.1: We say that the differentiable trajectory $y(t) \in Y$ is *admissible* if $p(y(t)) \cdot (dy(t)/dt) \geq 0$ for every t , with equality only if $y(t)$ is profit maximizing for $p(y(t))$ (in which case we could say that we are at a long-run equilibrium).

A difference between the price and the quantity tâtonnement approaches that adds appeal to the second is that feasibility is now insured at any t and that, as a result, we can interpret the dynamics as happening in real time.^{69,70}

Will an admissible trajectory necessarily take us to long-run equilibrium? We cannot really explore this matter here in any detail. As usual, the answer is “only

67. We could also look at the general case where both could be in disequilibrium; see, for example, Mas-Colell (1986).

68. There is no difficulty in considering several. Also, Y can be interpreted as an individual or as an aggregate production set.

69. Nonetheless, it is important to realize that, even then, this is not a fully dynamic model: The optimization problems of the consumers remain static and free of expectational feedbacks and firms follow naive, short-run rules of adjustment (in a more positive spirit one might call this *adaptive*, rather than naive, behavior). For an extensive analysis of market adjustment procedures in real time, see Fisher (1983).

70. The quantity dynamics of Definition 17.H.1 are reminiscent of Marshall (1920) and are often referred to as *Marshallian dynamics*, especially in a partial equilibrium context. In contrast, the price dynamics are frequently called *Walrasian dynamics*.

under special circumstances." A limited, but important example (it covers the short-run/long-run model of Section 10.F) is described in Proposition 17.H.2.

Proposition 17.H.2: If there is a single strictly convex consumer, then any admissible trajectory converges to the (unique) equilibrium.

Proof: Consider $u(y(t) + \omega)$ where $u(\cdot)$ and ω are respectively the utility function and the endowments of the consumer. The unique equilibrium production vector is the single production vector $\bar{y}$ that maximizes $u(y + \omega)$ on Y ; recall the one-consumer, one-firm example of Section 15.C.

The argument is much simpler if we assume that $u(\cdot)$ is differentiable. We claim that utility must then be increasing along any admissible trajectory. Indeed,

$$\begin{aligned} \frac{du(y(t) + \omega)}{dt} &= \nabla u(y(t) + \omega) \cdot \frac{dy(t)}{dt} \\ &= \mu(t)p(y(t)) \cdot \frac{dy(t)}{dt} > 0, \end{aligned}$$

with equality only at equilibrium. Here we have used the fact that at a short-run (interior) equilibrium, the price vector $p(y(t))$ weighted by the marginal utility of wealth $\mu(t)$ must be equal to the vector of marginal utilities of the consumer. Now, since utility is increasing, we must necessarily reach the production vector $\bar{y}$ at which utility is maximized in the feasible production set (i.e., the equilibrium). This is illustrated in Figure 17.H.3. (We are sidestepping minor technicalities: to proceed completely rigorously, we should argue that the dynamics cannot be so sluggish that we never reach the equilibrium. To do so we would need, strictly speaking, to strengthen slightly the concept of an admissible trajectory). ■

Note that the single consumer of Proposition 17.H.2 could be a (positive) representative consumer standing for a population of consumers.

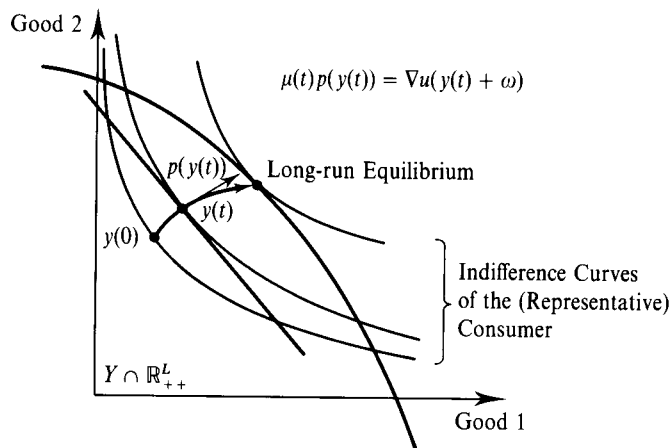


Figure 17.H.3
An example of
quantity tâtonnement.

17.1 Large Economies and Nonconvexities

As we have mentioned repeatedly, especially in Chapters 10 and 12, a central justification of the price-taking hypothesis is the assumption that every economic agent constitutes an insignificant part of the whole economy. Literally speaking, however, this cannot be satisfied in the model of this chapter because, formally, we allow for no more than a finite number I of consumers. (This is particularly true of our examples, where we typically have $I = 2$.) A straightforward reinterpretation is possible, however. We illustrate it for the case of a pure exchange economy.

Suppose we consider economies whose consumers have characteristics (preferences and endowments) that fall into I given types, with r consumers of each type (a generalization to unequal numbers per type is possible; see Exercise 17.1.1). That is, the set of consumers is formed by r replicas of a basic reference set of consumers. Furthermore, an allocation denoted by $(x_1, \dots, x_I)$ is understood now to specify that *each* consumer of type i consumes x_i (so the totality of consumers of type i consume rx_i). We observe then that the analysis and results presented up to this point are not modified by this reinterpretation; they simply do not depend in any way on the parameter r . In this way, we can conclude informally that the theory so far covers cases with an arbitrarily large number, even an infinity, of consumers; in particular, we see that any equilibrium of our earlier model is an equilibrium of the r -replica economy (for any integer $r \geq 1$).

There is, however, an important qualification. The ability to interpret the model and results in a manner that is fully independent of the number of consumers depends *crucially* on the convexity assumption on preferences. Without this assumption, it is not justified to neglect allocations that assign different consumption bundles to different consumers of the same type. Consider, for example, the Edgeworth box of Figure 17.1.1. If there is only one consumer of each type, then no equilibrium exists; but if we have two of each type, then there *is* an equilibrium. To see this, give ω_2 to the convex consumers, let one of the two nonconvex consumers receive the bundle x_1 , and let the other receive the different bundle x'_1 . Thus, in the nonconvex case, the

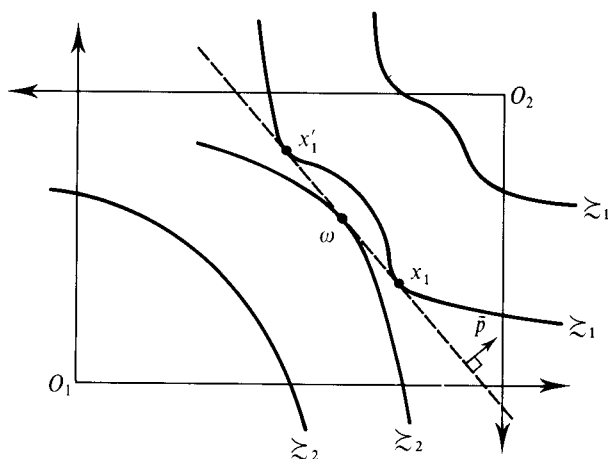


Figure 17.1.1

Equilibrium with nonconvex preferences in economies of changing size.

behavior of the economy can depend on the number of replicas: when we replicate an economy, new equilibria may emerge.⁷¹

The discussion of the previous example suggests an interesting observation: replication may actually *help* in the analysis of economies with nonconvexities, in the sense that an increase in the size of the economy (in terms of the number of replicas) may help insure existence of an equilibrium. Indeed, we devote the rest of this section to develop the argument that, if the economy is large enough, then the existence of an equilibrium is assured, or nearly so, even if preferences are not convex.⁷²

To see this, suppose that we have an exchange economy with I types. Consider a consumer of type i . If preferences are not convex (perhaps goods are indivisible), then the excess demand of this type is a correspondence $z_i(p)$. For $p \gg 0$, $z_i(p)$ is a compact set that may not be convex (as is the case for consumer 1 at $p = \bar{p}$ in Figure 17.I.1). Measuring in average, *per-replica*, terms, the excess demand correspondence of type i when there are r replicas is

$$\begin{aligned} z_{ir}(p) &= \frac{1}{r} (z_i(p) + \cdots + z_i(p)) \quad (\text{the sum has } r \text{ terms}) \\ &= \frac{1}{r} \{z_{i1} + \cdots + z_{ir} : z_{i1} \in z_i(p), \dots, z_{ir} \in z_i(p)\}. \end{aligned}$$

If we examine Figure 17.I.1 again, we see that, as $r \rightarrow \infty$, the set $z_{1r}(\bar{p}) + \{\omega_1\}$ fills the entire segment between the demand points x_1 and x'_1 . In particular, for any $\alpha \in [0, 1]$ and integer r we can find an integer $a_r \in [0, r]$ such that $|a_r/r - \alpha| \leq 1/r$ (note that the r numbers $\{1/r, \dots, r/r\}$ are evenly spaced in the interval $[0, 1]$). By putting a_r consumers at x_1 and $r - a_r$ consumers at x'_1 , we get a per-replica consumption of

$$\frac{a_r}{r} x_1 + \left(1 - \frac{a_r}{r}\right) x'_1 \in z_{ir}(p),$$

which, by taking r large enough, comes as close to $\alpha x_1 + (1 - \alpha)x'_1$ as we wish. It turns out that this convexifying property is completely general. For any $p \gg 0$ and whatever the number of commodities, as $r \rightarrow \infty$ the per-replica excess demand $z_{ir}(p)$ of type i converges as a set of the convex hull of $z_i(p)$. In the limit, the average per-replica excess demand correspondence of type i becomes $z_{i\infty}(p) = \text{convex hull } z_i(p)$. In the limit, therefore, the excess demand correspondence is convex valued and the existence of an equilibrium can be established as in Section 17.C.⁷³ In this sense,

71. That is, if we let $E(r)$ denote the equilibrium price set of the r replica economy, we have $E(1) \subset E(r)$, but the converse need not be true. Note, moreover, that for arbitrary $r'' > r' > 1$, there need not be any inclusion relationship between $E(r'')$ and $E(r')$ (except if $r'' = mr'$ for some integer $m > 1$, in which case $E(r') \subset E(r'')$).

72. See Starr (1969) for a classic contribution to this topic.

73. See the comment after the proof of Proposition 17.C.2 regarding demand correspondences, and also Exercise 17.C.1.

then, as r grows large, the economy must possess an allocation and price vector that constitute a “near” equilibrium.⁷⁴

In the previous reasoning, the convexification of aggregate excess demand, with its existence implication, depends on our ability to prescribe very carefully which of several indifferent consumptions each consumer has to choose. Only in this way can we make sure that the aggregate consumption will be precisely right. Whatever we may think about the possible processes that may lead consumers to select among indifferent optimal choices in the right proportions, there can be little doubt that it would be better if we did not have to worry about this; that is, if, given any price, practically every consumer had a single optimal choice. It is therefore of interest to point out that, while not a necessity, this is a most plausible occurrence if the number of consumers is large. Indeed, *if the distribution of individual preferences is dispersed across the population* (so that, in particular, no two consumers are exactly identical⁷⁵), *then even if the individual excess demands are true correspondences, the limit average may well be a (continuous) function.* This is because, at any p , only a vanishingly small proportion of consumers may display a nonconvexity at p . We commented on this point in Appendix A to Chapter 4 and we illustrate it further in Figure 17.I.2. Consider the Edgeworth box of Figure

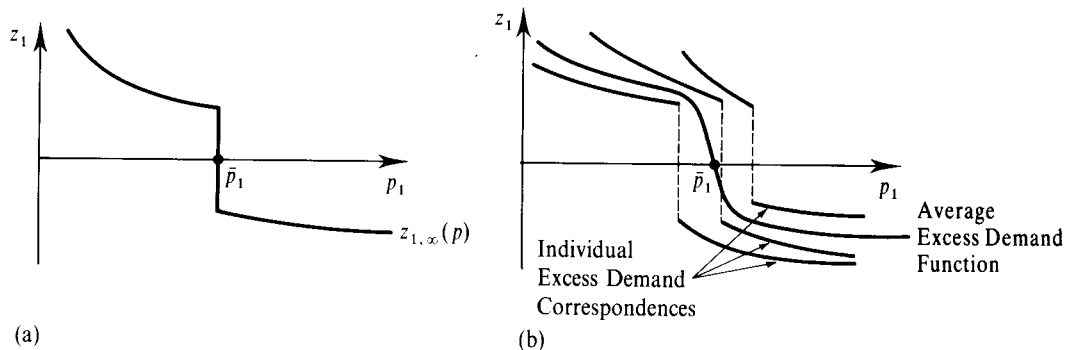


Figure 17.I.2 The aggregate demand from dispersed individual demand is a continuous function. (a) Aggregate excess demand for the Edgeworth box of Figure 17.I.1 with a continuum of consumers of each type (i.e., $r = \infty$). (b) Individual excess demands are dispersed.

17.I.1, but with a continuum of consumers of each type. Since in this economy all the consumers of type 1 are identical, all of them exhibit a “consumption switch” (a nonconvexity) at precisely the same $\bar{p}$. The excess demand correspondence for the first good in this economy is represented in Figure 17.I.2(a) as $z_{1,\infty}(\cdot)$.⁷⁶ But if tastes of type 1 consumers exhibit variation, even if slight, then we would expect that no significant group of consumers simultaneously switches at any p and therefore that, as in Figure 17.I.2(b), average demand will be well defined at any p and will change only gradually with p .

74. Roughly speaking, by a “near” equilibrium we mean an allocation and price vector that is close to satisfying the conditions of an equilibrium. A precise technical definition of this concept can be given, but we shall not do so here.

75. We note, as an incidental matter, that in the limit with an infinity of agents this requirement is incompatible with the existence of only a finite number of types. To deal with this case, the formal setting would need to be extended.

76. Precisely, $z_{1,\infty}(\cdot)$ is the correspondence whose graph is the limit graph, as r goes to ∞ , of the correspondences $z_r(\cdot)$ defined by $z_r(p) = (1/r)(z(p) + \cdots + z(p))$, where the sum has r terms and $z(\cdot)$ is the excess demand correspondence of the two-consumer economy in the Edgeworth box of Figure 17.I.1.

We comment briefly on economies with production. Suppose that the consumption side of the economy is generated, as before, as the r -replica of a basic reference set of (possibly nonconvex) consumers. There are also J production sets Y_j . Each Y_j is closed, contains the origin, and satisfies free disposal (these are all standard assumptions). In addition, we assume that there is an upper bound (a capacity bound perhaps) on every Y_j ; that is, there is a number s such that $y_{\ell j} \leq s$ for all ℓ and $y_j \in Y_j$. The production sets may be nonconvex.

It is then possible to argue that the economy will possess a near equilibrium if r is large relative to the bound s (i.e., if the size of the consumption side of the economy is large relative to the maximal size of a *single* firm). On the average, the production side of the economy is also being convexified, so to speak (see the small-type discussion of Section 5.E for a related point).⁷⁷ Note that the boundedness property of the production sets is important. Suppose, for example, that every firm has the technology represented in Figure 15.C.3. Then no matter how many consumers there are, the potential profits of every firm are infinite (as long as $p_2 > 0$). Thus, there is no reasonable sense in which a near equilibrium exists. For the averaging-out effect to work the nonconvexity in production has to be of bounded size (see Exercise 17.I.2).

APPENDIX A: CHARACTERIZING EQUILIBRIUM THROUGH WELFARE EQUATIONS

We have seen, beginning in Section 17.B, that if our economy satisfies sufficiently nice properties (e.g., strict convexity of preferences) then we can resort, for the purposes of the analysis, to formalizing our theory by means of highly reduced systems of equilibrium equations. In the text of this chapter, we have focused on excess demand equations. But this is not the only possibility. In this appendix, we briefly illustrate a second approach that builds on the welfare properties of equilibria.

We again concentrate on a pure exchange economy in which each consumer $i = 1, \dots, I$ has the consumption set $\mathbb{R}_+^L$ and continuous, strongly monotone, and strictly convex preferences. We also assume that $\omega_i \geq 0$ for all i and $\sum_i \omega_i \gg 0$.

We know from Chapter 16 that a Walrasian equilibrium of this economy is a Pareto optimum (Proposition 16.C.1). Therefore, to identify an equilibrium, we can as well restrict ourselves to Pareto optimal allocations. To this effect, suppose we fix continuous utility functions $u_i(\cdot)$ for the I consumers with $u_i(0) = 0$. Then to every vector $s = (s_1, \dots, s_I)$ in the simplex $\Delta = \{s' \in \mathbb{R}_+^L : \sum_i s'_i = 1\}$ we can associate a unique Pareto optimal allocation $x(s) \in \mathbb{R}_+^{LI}$ such that $(u_1(x_1(s)), \dots, u_I(x_I(s)))$ is

77. Observe that the average is with respect to r (the size of the economy in terms of the number of consumers), not with respect to J . If, as r increases, J is made to vary and is kept in some approximate fixed proportion with r , then from the qualitative point of view it does not matter how we measure size (this is a possible way to interpret, in the current context, the discussion of Section 5.E). But for the validity of the convexifying effect there is no need to vary J with r . The number J may be kept fixed and, thus, J could well be small relative to r (in which case the “averaged” economy is practically one of pure exchange) or it could be large; it could even be that $J = \infty$. The last case corresponds to a model with free entry, where the equilibrium—or the near equilibrium—determines, endogenously, the set of active firms. Typically, with free entry the set of the active firms increases as the number of consumers, measured by r , grows (this point has also been discussed in Section 10.F in a partial equilibrium context; there is not much more to add here).

APPENDIX B: A GENERAL APPROACH TO THE EXISTENCE OF WALRASIAN EQUILIBRIUM

The purpose of this appendix is to offer a treatment of the existence question at the level of generality of the model of Chapter 16. The results presented correspond roughly to those of Arrow and Debreu (1954) and McKenzie (1959).

As when dealing with the second welfare theorem in Section 16.D, and for exactly the same technical reasons, it is useful to concentrate on establishing the existence of a *Walrasian quasiequilibrium*. This is a weaker notion than Walrasian equilibrium in that consumers are required to maximize preferences only relative to consumptions that cost strictly less than the available amount of wealth.

Definition 17.BB.1: An allocation $(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*)$ and a price system $p \neq 0$ constitute a *Walrasian quasiequilibrium* if

- (i) For every j , $p \cdot y_j \leq p \cdot y_j^*$ for all $y_j \in Y_j$.
- (ii') For every i , $p \cdot x_i^* \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$, and

$$\text{if } x_i \succ_i x_i^* \text{ then } p \cdot x_i \geq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*.$$
- (iii) $\sum_i x_i^* = \sum_i \omega_i + \sum_j y_j^*$.

Definition 17.BB.1 is identical to Definition 17.B.1 of a Walrasian equilibrium except that the preference maximization condition (ii) of Definition 17.B.1 has been replaced by the weaker condition (ii'). With local nonsatiation, condition (ii') is equivalent to the requirement that x_i^* minimizes expenditure for the price vector p in the set $\{x_i \in X_i: x_i \succeq_i x_i^*\}$. The expenditure minimization problem has better continuity properties with respect to prices than does the preference maximization problem. Thus, in Figure 17.BB.1 we have $x_i^n = x_i^n(p^n, p^n \cdot x_i^n)$ and $p^n \cdot x_i^n = e(p^n, u_i(x_i^n))$; that is, x_i^n is preference maximizing relative to the price-wealth pair $(p^n, p^n \cdot x_i^n)$ and expenditure minimizing relative to the price-utility pair $(p^n, u_i(x_i^n))$. However, as $p^n \rightarrow p$ and $x_i^n \rightarrow \bar{x}_i$, we see that $\bar{x}_i$ fails to maximize $u_i(\cdot)$ relative to $(p, p \cdot \bar{x}_i)$, whereas $\bar{x}_i$ still minimizes expenditure relative to $(p, u_i(\bar{x}_i))$; that is, $p \cdot \bar{x}_i = e(p, u_i(\bar{x}_i))$. Because continuity is an important requirement for existence analysis, it is more convenient to prove the existence of a quasiequilibrium than that of an equilibrium. Ultimately, however, we are interested in equilibrium. As it turns out, it is relatively easy to

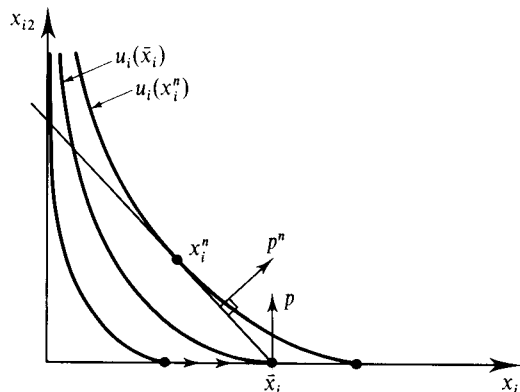


Figure 17.BB.1
Discontinuity of
preference
maximization.

state conditions implying that a quasiequilibrium is automatically an equilibrium. We devote the next few paragraphs to elaborating on this point.

We begin with Definition 17.BB.2

Definition 17.BB.2: The Walrasian quasiequilibrium (x^*, y^*, p) satisfies the *cheaper consumption condition for consumer i* if there is $x_i \in X_i$ such that $p \cdot x_i < p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$.

We then have Proposition 17.BB.1.

Proposition 17.BB.1: Suppose that consumption sets are convex and preferences are continuous. Then any consumer who at the Walrasian quasiequilibrium (x^*, y^*, p) satisfies the cheaper consumption condition must be preference maximizing in his budget set. Hence, if the cheaper consumption condition is satisfied for all i , (x^*, y^*, p) is also a Walrasian equilibrium.

Proof: Suppose that i satisfies the cheaper consumption condition; that is, there is $x_i \in X_i$ with $p \cdot x_i < p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$. If x_i^* fails to be preference maximizing, then there is x_i' such that $x_i' \succ_i x_i^*$ and x_i' is in the budget set of consumer i . Denote $x_i^n = (1 - (1/n))x_i' + (1/n)x_i$. Then $x_i^n \in X_i$ and $p \cdot x_i^n < p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$ for all n , and $x_i^n \rightarrow x_i'$ as $n \rightarrow \infty$. By continuity of preferences, for large enough n we will have $x_i^n \succ_i x_i^*$. But then consumer i violates condition (ii') of the definition of quasiequilibrium. ■

Suppose that, for every j , $0 \in Y_j$, and for every i , X_i is convex and $\omega_i \geq \hat{x}_i$ for some $\hat{x}_i \in X_i$. Suppose, in addition, that the weak condition " $\sum_i \omega_i + \sum_j \hat{y}_j \gg \sum_i \hat{x}_i$ for some $(\hat{y}_1, \dots, \hat{y}_j) \in Y_1 \times \dots \times Y_j$ " is satisfied. Then the quasiequilibrium (x^*, y^*, p) satisfies the cheaper consumption condition for consumer i in, for example, either of the following situations (Exercise 17.BB.1):

- (a) $p \geq 0$, $p \neq 0$, and $\omega_i \gg \hat{x}_i$,
- (b) $p \gg 0$ and $\omega_i \neq \hat{x}_i$.

To have $p \geq 0$ at quasiequilibrium it suffices that one production set satisfies the free-disposal condition. Guaranteeing $p \gg 0$ is more difficult. It will happen if, for every i , $X_i + \mathbb{R}_+^L \subset X_i$ and preferences are continuous and strongly monotone. To see this, note that by the monotonicity of preferences and the expenditure-minimization property of quasiequilibrium we must have $p \geq 0$, $p \neq 0$, at any quasiequilibrium. It follows that $p \cdot (\sum_i \omega_i + \sum_j y_j^*) \geq p \cdot (\sum_i \omega_i + \sum_j \hat{y}_j) > p \cdot (\sum_i \hat{x}_i)$. Hence, there is at least one consumer with wealth larger than $p \cdot \hat{x}_i$. But this consumer must be maximizing preferences (by Proposition 17.BB.1), which, by the strong monotonicity property, can only occur if every price is positive (i.e., if no good is free).

Although convenient, neither condition (a) nor (b) can be regarded as extremely weak. It would be unfortunate if the validity of the theory were restricted to them. But this is not so: much weaker conditions are available. In particular, McKenzie (1959) has developed a theory of *indecomposable* economies that guarantees that at a quasiequilibrium the cheaper consumption condition is satisfied for every consumer (and therefore the quasiequilibrium is an equilibrium). The basic idea, informally described, is that an economy is indecomposable if, no matter how we partition the economy into two groups, each of the groups has something for which the other group is willing to exchange something of its own. (See Exercise 17.BB.2.)

We now turn to the existence of a Walrasian quasiequilibrium. The aim is to establish the general existence result in Proposition 17.BB.2.

Proposition 17.BB.2: Suppose that for an economy with $I > 0$ consumers and $J > 0$ firms we have

- (i) For every i ,
 - (i.1) $X_i \subset \mathbb{R}^L$ is closed and convex;
 - (i.2) $\succeq_i$ is a rational, continuous, locally nonsatiated, and convex preference relation defined on X_i ;
 - (i.3) $\omega_i \geq \hat{x}_i$ for some $\hat{x}_i \in X_i$.
- (ii) Every $Y_j \subset \mathbb{R}^L$ is closed, convex, includes the origin, and satisfies the free-disposal property.
- (iii) The set of feasible allocations

$$A = \{(x, y) \in \mathbb{R}^{LI} \times \mathbb{R}^{LJ} : x_i \in X_i \text{ for all } i, y_j \in Y_j \text{ for all } j, \text{ and}$$

$$\sum_i x_i \leq \sum_i \omega_i + \sum_j y_j\}$$

is compact.

Then a Walrasian quasiequilibrium exists.

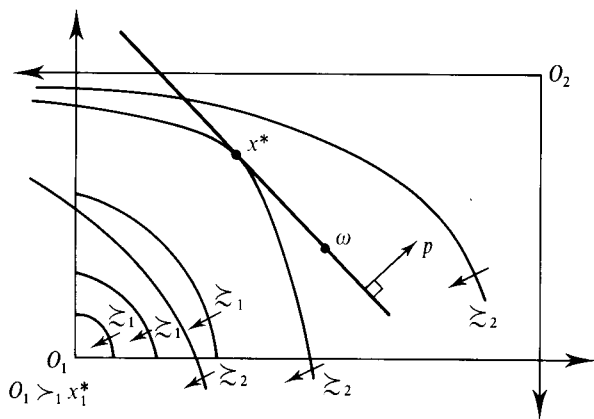
We comment briefly on the assumptions. As we have repeatedly illustrated (in Chapters 10, 15, and 16), the convexity assumptions on individual preferences and technologies cannot be dispensed with.⁷⁹ The Edgeworth box example of Figure 17.BB.2 shows that the local nonsatiation condition is also required.⁸⁰ In contrast, the assumption of rational preferences is entirely dispensable (see the comments at the end of this appendix). The free disposal condition $-\mathbb{R}_+^L \subset Y_j$ is also only a matter of convenience.⁸¹ Allowing for negative prices, we could simply drop it from our list of conditions (see Exercise 17.BB.3). The assumption (i.3) says that ω_i may not belong to the consumption set but that it is possible to reach the consumption set by just eliminating some amounts of commodities from ω_i .⁸² Finally, in Appendix A to Chapter 16 we have already investigated the conditions under which the set of feasible allocations is compact.

79. Recall, however, the important qualification of Section 17.I, and see also the discussion at the end of this appendix.

80. In Figure 17.BB.2, the second consumer has conventional strongly monotone preferences; but for the first consumer both commodities are bads and, therefore, he is satiated at the origin. Also $\omega_1 \gg 0$ and $\omega_2 \gg 0$. Suppose that $x^* = (x_1^*, x_2^*)$ and price vector $p \neq 0$ constitute a Walrasian quasiequilibrium. Because the preferences of the second consumer are strongly monotone, we must have $p \gg 0$. By profit maximization (using the free-disposal technology) and the possibility of inaction, we have $p \cdot (x_1^* + x_2^* - \omega_1 - \omega_2) \geq 0$. Since $p \cdot x_2^* \leq p \cdot \omega_2$, this yields $p \cdot x_1^* \geq p \cdot \omega_1 > 0$. But then (x^*, p) cannot be a Walrasian quasiequilibrium because consuming nothing costs zero and is preferred by the first consumer to any other consumption.

81. Because Y_j is convex and closed, $-\mathbb{R}_+^L \subset Y_j$ implies $Y_j - \mathbb{R}_+^L \subset Y_j$ (Exercise 5.B.5).

82. A stronger condition would require that $\omega_i \gg \hat{x}_i$ for every i . With this assumption, Proposition 17.BB.2 yields the existence of a true equilibrium, not just of a quasiequilibrium. Economically, however, the latter assumption is considerably stronger: $\omega_i \geq \hat{x}_i$ can be interpreted (keeping in mind the possibility of free disposal) simply as saying that consumer i could survive without entering the markets of the economy, while $\omega_i \gg \hat{x}_i$ says that the consumer can supply to the market a strictly positive amount of every good.

**Figure 17.BB.2**

Equilibrium does not exist: the preferences of the first consumer are satiated.

Proof of Proposition 17.BB.2

The approach we follow takes advantage of the fact that the reader may already have been exposed in Chapter 8 to the notion of the Nash equilibrium of a normal form game and, more particularly, to the existence results for Nash equilibrium using best-response correspondences contained in Appendix A to Chapter 8. A game-theoretic approach to the existence of Walrasian equilibrium was taken in the classic paper of Arrow and Debreu (1954). Here we follow Gale and Mas-Colell (1975).

Definition 17.BB.3: An allocation (x^*, y^*) and a price system $p \neq 0$ constitute a *free-disposal quasiequilibrium* if

- (i) for every j , $p \cdot y_j \leq p \cdot y_j^*$ for all $y_j \in Y_j$.
- (ii') For every i , $p \cdot x_i^* \leq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$, and

$$\text{if } x_i \succ_i x_i^* \text{ then } p \cdot x_i \geq p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*.$$

- (iii') $\sum_i x_i^* \leq \sum_i \omega_i + \sum_j y_j^*$ and $p \cdot (\sum_i x_i^* - \sum_i \omega_i - \sum_j y_j^*) = 0$.

Thus, all we have done is replace in Definition 17.BB.1 of a quasiequilibrium the exact feasibility condition " $\sum_i x_i^* = \sum_i \omega_i + \sum_j y_j^*$ " by (iii') above. That is, we allow the excess supply of some goods provided that they are free. In Exercise 17.BB.4 you are asked to show that if one production set, say Y_1 , satisfies the free-disposal property and if $(x_1^*, \dots, x_I^*, y_1^*, \dots, y_J^*, p)$ is a free-disposal quasiequilibrium, then there is $y_1'^* \leq y_1^*$ such that $(x_1^*, \dots, x_I^*, y_1'^*, y_2^*, \dots, y_J^*, p)$ is a Walrasian quasiequilibrium. Therefore, to establish Proposition 17.BB.2, it is enough for us to show that a free-disposal quasiequilibrium exists.

We proceed to formalize the free-disposal quasiequilibrium notion as a kind of noncooperative equilibrium for a certain game among $I + J + 1$ players. The I and J players are the consumers and the firms, respectively, and their strategies are demand–supply vectors. The extra player is a fictitious *market agent* (a “grand coordinator”) having as his strategy the prices of the L different goods.

Since the set A of feasible allocations is bounded, there is $r > 0$ such that whenever $(x_1, \dots, x_I, y_1, \dots, y_J) \in A$ we have $|x_{\ell i}| < r$ and $|y_{\ell j}| < r$ for all i, j , and ℓ . Because we need to have compactness of strategy sets to establish existence, we begin by

replacing every X_i and every Y_j by a truncated version:

$$\hat{X}_i = \{x_i \in X_i : |x_{\ell i}| \leq r \text{ for all } \ell\},$$

$$\hat{Y}_j = \{y_j \in Y_j : |y_{\ell j}| \leq r \text{ for all } \ell\}.$$

Note that $A \subset \hat{X}_1 \times \cdots \times \hat{X}_I \times \hat{Y}_1 \times \cdots \times \hat{Y}_J$. Because $(\hat{x}_1, \dots, \hat{x}_I, \dots, 0, \dots, 0) \in A$, it follows that $\hat{x}_i \in \hat{X}_i$ for every i , and $0 \in \hat{Y}_j$ for every j . In particular, all the strategy sets are nonempty. Lemma 17.BB.1 shows that in our search for a free-disposal quasi-equilibrium we can limit ourselves to the truncated economy.

Lemma 17.BB.1: If all X_i and Y_j are convex and (x^*, y^*, p) is a free-disposal quasi-equilibrium in the truncated economy, that is, if (x^*, y^*, p) satisfies Definition 17.BB.3 of free-disposal quasiequilibrium with the consumption and production sets replaced by their truncated versions, then (x^*, y^*, p) is also a free-disposal quasiequilibrium for the original untruncated economy.

Proof of Lemma 17.BB.1: Consider a consumer i (the reasoning is similar for a firm). Because $(x^*, y^*) \in A$, we have $|x_{\ell i}^*| < r$ for all ℓ ; that is, the consumption bundle of consumer i is interior to the truncation bound. Suppose now that x_i^* fails to satisfy condition (ii') of Definition 17.BB.3 in the nontruncated economy, that is, that there is an $x_i \in X_i$ such that $x_i \succ_i x_i^*$, and $p \cdot x_i < p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$. Denote $x_i^n = (1 - (1/n))x_i^* + (1/n)x_i$. For all n we have $p \cdot x_i^n < p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$ and, by the convexity of preferences, $x_i^n \succeq_i x_i^*$. Also, we can choose an n large enough to have $|x_{\ell i}^n| < r$ for all ℓ . By local nonsatiation there must then be an $x_i' \in \hat{X}_i$ such that $x_i' \succ_i x_i^n$ and $p \cdot x_i' < p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$. But then $x_i' \in \hat{X}_i$ and $x_i' \succ_i x_i^n \succeq_i x_i^*$, and so in the truncated economy x_i^* fails to satisfy condition (ii') of Definition 17.BB.3. Thus, (x^*, y^*, p) must not be a free-disposal quasiequilibrium in the truncated economy. This contradiction establishes the result. ■

We are now ready to set up a simultaneous-move noncooperative game. To do so we need to specify the players' strategy sets and payoff functions. To simplify notation we assign to every consumer i , price vector p and production profile $y = (y_1, \dots, y_J)$, a limited liability amount of wealth

$$w_i(p, y) = p \cdot \omega_i + \text{Max} \left\{ 0, \sum_j \theta_{ij} p \cdot y_j \right\}.$$

The strategy sets are:

For consumer i : $\hat{X}_i$

For firm j : $\hat{Y}_j$

For the market agent: $\Delta = \{p \in \mathbb{R}^L : p_\ell \geq 0 \text{ for all } \ell \text{ and } \sum_\ell p_\ell = 1\}$.

Given a strategy profile $(x, y, p) = (x_1, \dots, x_I, y_1, \dots, y_J, p)$, the payoff functions and best-responses of the different agents are:

Consumer i : Chooses consumption vectors $x_i' \in \hat{X}_i$ such that
 (1) $p \cdot x_i' \leq w_i(p, y)$ and
 (2) $x_i' \succeq_i x_i''$ for all $x_i'' \in \hat{X}_i$ satisfying $p \cdot x_i'' < w_i(p, y)$.
 (Consumer i 's payoff function can be thought of as giving a payoff 1 if he chooses a consumption vector satisfying this condition, and 0 otherwise.)

Denote by $\tilde{x}_i(x, y, p) \subset \hat{X}_i$ the set of consumption bundles x'_i so defined.

Firm j : Chooses productions $y'_j \in \hat{Y}_j$ that are profit maximizing for p on $\hat{Y}_j$. (Firm j 's payoff function is simply its profit.)

Denote by $\tilde{y}_j(x, y, p) \subset \hat{Y}_j$ the set of production plans y'_j so defined.

Market Agent: Chooses prices $q \in \Delta$ so as to solve

$$\text{Max}_{q \in \Delta} \left(\sum_i x_i - \sum_i \omega_i - \sum_j y_j \right) \cdot q. \quad (17.BB.1)$$

Denote by $\tilde{p}(x, y, p)$ the set of price vectors q so defined.

Only the behavior of the market agent needs comment. Given the total excess demand vector, the market agent chooses prices so as to maximize the value of this vector. Hence, he puts the whole weight of prices (which, recall, have been normalized to lie in the unit simplex) into the commodities with maximal excess demand. As we have already observed when doing the same thing in the proof of Proposition 17.C.1, this is in accord with economic logic: if the objective is to eliminate the excess demand of some commodities, try raising their prices as much as possible.

Lemma 17.BB.2 says that an equilibrium of this noncooperative game yields a free-disposal quasiequilibrium for the truncated economy.

Lemma 17.BB.2: Suppose that (x^*, y^*, p) is such that $x_i^* \in \tilde{x}_i(x^*, y^*, p)$ for all i , $y_j^* \in \tilde{y}_j(x^*, y^*, p)$ for all j , and $p \in \tilde{p}(x^*, y^*, p)$. Then (x^*, y^*, p) is a free-disposal quasiequilibrium for the truncated economy.

Proof of Lemma 17.BB.2: We note first that $p \cdot y_j^* \geq 0$ for every j (because $0 \in \hat{Y}_j$). By the definition of $\tilde{x}_i(\cdot)$ and $\tilde{y}_j(\cdot)$, conditions (i) and (ii') of Definition 17.BB.3 are then automatically satisfied. Hence, the only property that remains to be established is (iii'), that is,

$$\sum_i x_i^* - \sum_i \omega_i - \sum_j y_j^* \leq 0 \quad \text{and} \quad p \cdot \left(\sum_i x_i^* - \sum_i \omega_i - \sum_j y_j^* \right) = 0.$$

We have $p \cdot x_i^* \leq w_i(p, y^*) = p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$ for all i and therefore

$$p \cdot \left(\sum_i x_i^* - \sum_i \omega_i - \sum_j y_j^* \right) \leq 0.$$

This implies $\sum_i x_i^* - \sum_i \omega_i - \sum_j y_j^* \leq 0$ because otherwise the value of the solution to problem (17.BB.1) would be positive and so p (which as we have just seen has $p \cdot (\sum_i x_i^* - \sum_i \omega_i - \sum_j y_j^*) \leq 0$) could not be a maximizing solution vector, that is, a member of $\tilde{p}(x^*, y^*, p)$. It follows that $(x^*, y^*) \in A$ and so, $x_i^* < r$ for all i and ℓ . From this we get that the budget equations are satisfied with equality (i.e., $p \cdot x_i^* = p \cdot \omega_i + \sum_j \theta_{ij} p \cdot y_j^*$ for all i) because otherwise local nonsatiation yields that for some consumer i there is a preferred consumption strictly interior to consumer i 's budget set in the truncated economy, implying $x_i^* \notin \tilde{x}_i(x^*, y^*, p)$. We therefore conclude that we also have $p \cdot (\sum_i x_i^* - \sum_i \omega_i - \sum_j y_j^*) = 0$. This completes the proof. ■

Now, as we discussed in Appendix A to Chapter 8 (see the proof of Proposition 8.D.3 presented there), under appropriate conditions on the best-response correspondences, this noncooperative game has an equilibrium.

Lemma 17.BB.3: Suppose that the correspondences $\tilde{x}_i(\cdot)$, $\tilde{y}_j(\cdot)$, and $\tilde{p}(\cdot)$ are nonempty, convex valued, and upper hemicontinuous. Then there is (x^*, y^*, p) such that $x_i^* \in \tilde{x}_i(x^*, y^*, p)$ for all i , $y_j^* \in \tilde{y}_j(x^*, y^*, p)$ for all j , and $p \in \tilde{p}(x^*, y^*, p)$.

Proof of Lemma 17.BB.3: We are simply looking for a fixed point of the correspondence $\Psi(\cdot)$ from $X_1 \times \cdots \times X_I \times Y_1 \times \cdots \times Y_J \times \Delta$ to itself defined by

$$\Psi(x, y, p) = \tilde{x}_1(x, y, p) \times \cdots \times \tilde{x}_I(x, y, p) \times \tilde{y}_1(x, y, p) \times \cdots \times \tilde{y}_J(x, y, p) \times \tilde{p}(x, y, p).$$

The correspondence $\Psi(\cdot)$ is nonempty, convex valued, and upper hemicontinuous. The existence of a fixed point follows directly from Kakutani's fixed point theorem (see Section M.I of the Mathematical Appendix). ■

Lemmas 17.BB.4 to 17.BB.6 verify that the best-response correspondence of this noncooperative game is nonempty, convex valued, and upper hemicontinuous.⁸³

Lemma 17.BB.4: For all strategy profiles (x, y, p) , the sets $\tilde{x}_i(x, y, p)$, $\tilde{y}_j(x, y, p)$, and $\tilde{p}(x, y, p)$ are nonempty.

Proof of Lemma 17.BB.4: For $\tilde{y}_j(x, y, p)$ and $\tilde{p}(x, y, p)$ the claim is clear enough since we are maximizing a continuous (in fact, linear) function on, respectively, the nonempty, compact sets $\hat{Y}_j$ and Δ . For $\tilde{x}_i(x, y, p)$, recall that the continuity of $\succeq_i$ implies the existence of a continuous utility representation $u_i(\cdot)$ for $\succeq_i$.⁸⁴ Let x'_i be a maximizer of the continuous function $u_i(x_i)$ on the nonempty compact budget set $\{x_i \in \hat{X}_i : p \cdot x_i \leq w_i(p, y^*)\}$. Then $x'_i \in \tilde{x}_i(x, y, p)$. The budget set is nonempty because $\hat{x}_i \in \hat{X}_i$ and $\hat{x}_i \leq \omega_i$. With $p \geq 0$, this implies that $p \cdot \hat{x}_i \leq p \cdot w_i \leq w_i(p, y^*)$. ■

Lemma 17.BB.5: For all strategy profiles the sets $\tilde{x}_i(x, y, p)$, $\tilde{y}_j(x, y, p)$, and $\tilde{p}(x, y, p)$ are convex.

Proof of Lemma 17.BB.5: We establish the claim for $x_i(x, y, p)$. You are asked to complete the proof in Exercise 17.BB.6.

Suppose that $x_i, x'_i \in \tilde{x}_i(x, y, p)$ and consider $x_{i\alpha} = \alpha x_i + (1 - \alpha)x'_i$, for any $\alpha \in [0, 1]$. Note first that $p \cdot x_{i\alpha} \leq w_i(p, y)$. In addition, by the convexity of preferences we cannot have $x_i \succ_i x_{i\alpha}$ and $x'_i \succ_i x_{i\alpha}$ (Exercise 17.BB.5). So suppose that $x_{i\alpha} \succeq_i x_i$. Consider now any $x''_i \in \hat{X}_i$ with $p \cdot x''_i < w_i(p, y)$. Then since $x_i \in \tilde{x}_i(x, y, p)$ we have $x_i \succeq_i x''_i$, and so $x_{i\alpha} \succeq_i x''_i$. We conclude that $x_{i\alpha} \in \tilde{x}_i(x, y, p)$. A similar conclusion follows if $x_{i\alpha} \succeq_i x'_i$. Hence, $\tilde{x}_i(x, y, p)$ is a convex set. ■

Lemma 17.BB.6: The correspondences $\tilde{x}_i(\cdot)$, $\tilde{y}_j(\cdot)$, and $\tilde{p}(\cdot)$ are upper hemicontinuous.

Proof of Lemma 17.BB.6: Again, we limit ourselves to $\tilde{x}_i(\cdot)$. Exercise 17.BB.7 asks you to complete the proof for $\tilde{y}_j(\cdot)$ and $\tilde{p}(\cdot)$.

83. For the firms and the market game this result is covered by Proposition 8.D.3, but for the consumers we need a special argument (as defined, the payoff functions of the consumers are not continuous).

84. This was proved in Proposition 3.C.1 for monotone preferences on $\mathbb{R}_+^L$. As we pointed out there, however, the conclusion actually depends only on the continuity of the preference relation.

Let $p^n \rightarrow p$, $y^n \rightarrow y$, $x^n \rightarrow x$, and $x_i'^n \rightarrow x_i'$ as $n \rightarrow \infty$, and suppose that $x_i' \in \tilde{x}_i(x^n, y^n, p^n)$. We need to show that $x_i' \in \tilde{x}_i(p, x, y)$.

From $p^n \cdot x_i'^n \leq w_i(p^n, y^n)$ we get $p \cdot x_i' \leq w_i(p, y)$. Consider now any $x_i'' \in \hat{X}_i$ with $x_i'' \succ_i x_i'$. Then, by the continuity of preferences, $x_i'' \succ_i x_i'^n$ for n large enough. Hence, $p^n \cdot x_i'' \geq w_i(p^n, y^n)$. Going to the limit we get $p \cdot x_i'' \geq w_i(p, y)$. Thus, we conclude that, as we wanted, $x_i' \in \tilde{x}_i(x, y, p)$. It is, incidentally, because of the need to establish this closed-graph property that we have replaced preference maximization by the weaker objective of expenditure minimization in the definition of the objectives of the consumer. ■

The combination of Lemmas 17.BB.4 to 17.BB.6 establishes that the given best-response correspondences satisfy the properties required in Lemma 17.BB.3 for the existence of a fixed-point, which completes the proof of Proposition 17.BB.2. ■

The assumptions on preferences and technologies can be weakened in an important respect. Our existence argument requires only that the best-response correspondence $\tilde{x}_i(x, y, p)$ and $\tilde{y}_j(x, y, p)$ be convex valued and upper hemicontinuous. Beyond this, the proof imposes no restrictions whatsoever on the dependence of consumers' and firms' choices on the "state" variables (x, y, p) . Thus we could allow consumers' tastes, or firms' technologies, to depend on prices (money illusion?), on the choices of other consumers or firms (a form of externalities), or even on own consumption (e.g., tastes could depend on a current reference point—a source of incompleteness or nontransitivity of preferences already illustrated in Chapter 1).^{85,86} The following is an example of the sort of generality that can be accommodated: Suppose that consumer preferences are given to us by means of utility functions $u_i(\cdot; x, y, p)$ defined on X_i but dependent, in principle, on the state of the economy. If for every (x, y, p) the conditions of Proposition 17.BB.2 are satisfied, and the parametric dependence on (x, y, p) is continuous, then a Walrasian quasiequilibrium still exists. The proof does not need any change. We can make a similar point with respect to the possibility that firms' technologies depend on external effects, with, then, an added theoretical payoff. It allows us to see that equilibrium exists if the technology of the firm is convex: *it does not matter if the "aggregate" technology of the economy is convex*. See Exercise 17.BB.8 for more on this.

The existence proof we have given in this appendix is an example of a "large space" proof. The fixed-point argument (in our case phrased as a Nash equilibrium existence argument) has been developed in a disaggregated domain where all the equilibrating variables have been listed separately. The advantage of proceeding this way is that the argument remains very flexible and allows us to incorporate the weakest possible conditions without extra effort (as the last paragraph has illustrated). The disadvantage, of course, is that the fixed point may be

85. Suppose, for example, that the utility function of a consumer is given to us in the form $u_i(\cdot; x_i)$; that is, the evaluation of possible consumptions depends on the current consumption. Without loss of generality we can normalize $u_i(x_i; x_i) = 0$ for every x_i . Define the induced weak and strict preference relations $\succeq_i$ and $\succ_i$ on X_i by, respectively, " $x_i' \succeq_i x_i$ if $u_i(x_i'; x_i) \geq 0$ " and " $x_i' \succ_i x_i$ if $u_i(x_i'; x_i) > 0$." Then the relations $\succeq_i$ and $\succ_i$ contain all the relevant information for equilibrium analysis. Note, however, that it is perfectly possible for $\succeq_i$ not to be complete and for neither $\succeq_i$ nor $\succ_i$ to be transitive. See Shafer (1974) and Gale and Mas-Colell (1975) for more on this.

86. Another example of dependence on the overall consumption vector of the economy arises if, for example, we are considering equilibrium at a given point in time. Then current consumptions in the economy (e.g., purchases of physical or financial assets) will typically affect future prices; these, in turn, will influence current preferences via expectations.